Partisan Differences in Health Behaviors During the COVID-19 Pandemic

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Abstract

This study examines partisan differences in health behaviors during the COVID-19 pandemic using data from the Berkeley Interpersonal Contact Study (BICS). We analyze how political affiliation shaped individual responses to the pandemic, including social contact patterns, mask-wearing behavior, and vaccination uptake. Our findings reveal substantial and persistent partisan gaps in protective health behaviors that varied across geographic contexts and evolved over the course of the pandemic. These results have important implications for understanding how political polarization influences public health outcomes and for designing effective health communication strategies.

1 Introduction

The COVID-19 pandemic revealed a striking and troubling pattern in American public health: mortality rates diverged sharply along partisan lines. Counties that voted heavily for Donald Trump in 2020 experienced substantially higher COVID-19 death rates than those that supported Joe Biden [CITE: Gollwitzer, Krieger, etc.]. This partisan mortality divide was not a statistical artifact or a reflection of pre-existing health disparities alone—it emerged during the pandemic and widened over time, particularly after vaccines became available [CITE]. Early in the pandemic, mortality was higher in Democratic-leaning counties—reflecting the initial concentration of cases in urban areas like New York City. But by fall 2020, this pattern reversed, and Republican-leaning counties experienced persistently higher mortality rates through the remainder of the pandemic. The question is: why?

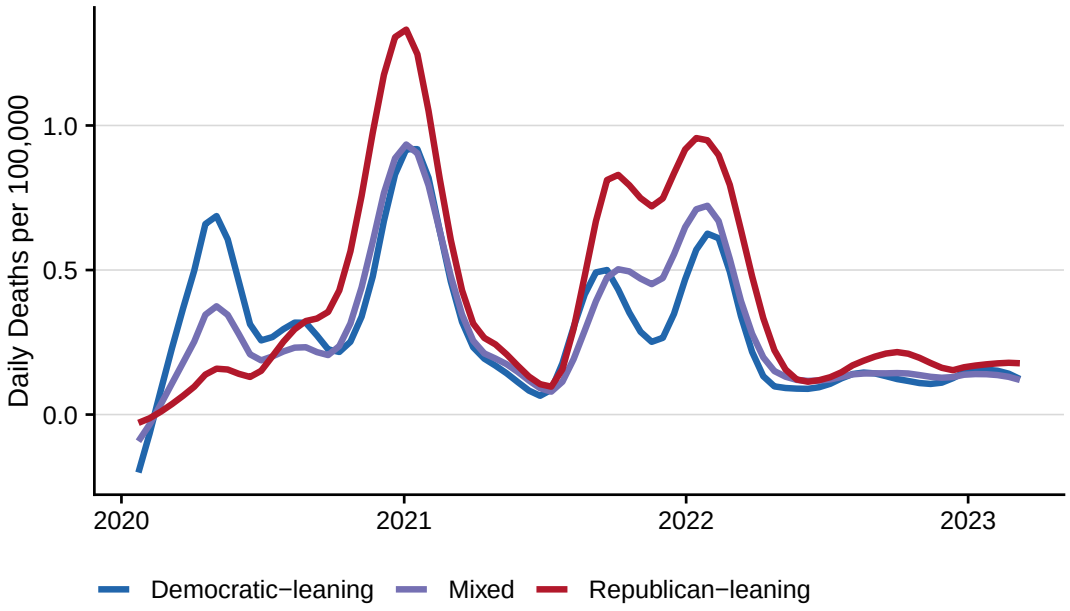

Figure 1: COVID-19 Mortality by County Partisan Composition

These aggregate patterns raise the possibility that individual partisans differed in their behavioral responses to the pandemic—that Republicans were less likely to reduce contacts, wear masks, and get vaccinated than Democrats. But county-level data alone cannot tell us whether such individual-level differences exist, how large they are, or whether they would persist if we accounted for the many other ways that Republican-leaning and Democratic-leaning places differ.

Two broad explanations have been offered for these partisan disparities. The first emphasizes policy environments: Democrats were concentrated in urban areas governed by Democratic officials who implemented stricter lockdowns, mask mandates, and business closures, while Republicans lived in areas with fewer restrictions. On this account, mortality differences reflect differential exposure to the virus driven by policy, not genuine differences in individual behavior. The second explanation emphasizes behavior: partisans may have genuinely differed in how they responded to the pandemic—in their willingness to reduce social contacts, wear masks, and get vaccinated—independent of the policies governing their communities. Adjudicating between these explanations is critical for understanding the role of political identity in shaping health outcomes and for designing effective public health responses to future crises.

There is strong theoretical reason to expect that partisan identity could shape health behavior directly. Over the past several decades, party identification in the United States has evolved from a summary of policy preferences into a fundamental social identity [CITE: Mason, Iyengar]. Partisanship now rivals race, religion, and ethnicity in its psychological salience and its capacity to structure social life [CITE]. Americans increasingly select friends, neighbors, and romantic partners based on party affiliation; they express greater social distance from and animosity toward members of the opposing party; and they view political events through the lens of partisan group membership rather than ideological principle [CITE: affective polarization literature]. This transformation of partisanship from preference to identity has profound implications for behavior. Social identities do not merely correlate with behavior—they drive it, through mechanisms including in-group conformity, social pressure, and responsiveness to elite cues from group leaders [CITE].

These dynamics of partisan identity found fertile ground in the domain of science and public health. Trust in scientific institutions had already been diverging along party lines for over a decade before COVID-19, with some scholars tracing the origins of this divide to the late 2000s [CITE]. But the pandemic dramatically accelerated this trend [CITE]. Questions that epidemiologists viewed as empirical—How severe is the virus? Do masks reduce transmission? Are vaccines safe and effective?— became markers of partisan allegiance [CITE]. Republican political leaders and conservative media figures frequently downplayed the severity of the pandemic, questioned the efficacy of masks, and expressed skepticism about vaccines, while Democratic leaders emphasized the threat and urged compliance with public health guidance [CITE]. If partisanship functions as a social identity, and if health behaviors became symbols of that identity, we should expect partisans to have diverged in their behavioral responses to the pandemic—not merely because they lived in different policy environments, but because their social identities motivated different responses to the same threat.

Prior research has documented substantial partisan differences in pandemic-related attitudes and some behaviors [CITE: Gadarian, Allcott, etc.]. However, this literature leaves critical questions unanswered. Most studies cannot distinguish whether observed partisan differences

reflect genuine behavioral divergence or simply the correlation between partisanship and the policy environments, geographic contexts, and demographic compositions of the places where partisans live. Moreover, we know little about the conditions under which partisan gaps in health behavior are largest or smallest. Before we can explain *why* partisan identity shapes health behavior, we must first establish *whether* it does so robustly—independent of policy, place, and demography—and *under what conditions* partisan identity matters most.

This paper uses data from the Berkeley Interpersonal Contact Study (BICS), a nationally representative survey that collected detailed information on social contact patterns, mask-wearing, and vaccination during the COVID-19 pandemic, to address these gaps. A key strength of the BICS data is that respondents reported the actual number of interpersonal contacts they had on the day prior to the survey, rather than responding to abstract categorical questions about social behavior. Contact patterns are a primary driver of infectious disease transmission [CITE: contact tracing literature, Mossong et al.], and having concrete counts—rather than ordinal self-assessments—provides a more direct and granular measure of the behaviors most relevant to disease spread. Respondents also reported whether they wore masks during each contact, providing a second behavioral measure with direct relevance to transmission risk [CITE]. Crucially, the survey also captured respondents' stated perceptions of pandemic severity; this allows us to examine whether attitudes and behaviors move together or diverge, a key test of whether expressed concern reflects genuine risk assessment or partisan signaling. The structure of these data makes the findings from this study relevant not only to researchers studying polarization and health behaviors, but also to those developing disease transmission models—which require precisely this type of contact and protective behavior data as inputs [CITE]. Moreover, understanding heterogeneity in contact patterns across subpopulations is critical for accurate disease modeling; research has shown that population-level averages can obscure important variation, and that heterogeneity in behavior can substantially shift disease dynamics even when overall means remain unchanged [CITE: heterogeneity in disease modeling literature]. Our approach is to examine whether partisan differences in health behavior persist across a range of contextual conditions, using patterns of variation to adjudicate between competing theoretical explanations. Although we cannot directly test causal mechanisms, we can ask: which theories survive

contact with the empirical patterns?

We exploit variation across policy environments, demographic composition, local partisan context, perceived severity, and pandemic conditions to ask whether partisan gaps narrow where competing theories predict they should. If differences are artifacts of policy or place, they should dissolve within comparable environments; if they reflect demographic sorting, they should close after controlling for covariates; if information and perception fully drive behavior, they should disappear among partisans who assess risk similarly; and if objective severity overrides identity, they should narrow when mortality is highest. The stakes of adjudication extend beyond academic debate: if partisan gaps are reducible to policy, place, or information, the remedies are institutional; if identity drives behavior independently of these factors, polarization may prove more durable, persisting even as policies converge and information environments shift. We refer to this latter possibility as "portable polarization"—the idea that partisans carry their behavioral patterns with them regardless of where they live or what surrounds them, implying that geographic mixing or mandate enforcement alone will not close the gap.

Several theoretical perspectives generate distinct predictions about how partisan gaps should vary across contexts. If partisan differences are primarily artifacts of policy environments, gaps should disappear when we compare partisans living under similar policies—within strict mandate areas, within lenient areas, and within areas with no mandates. If partisan differences reflect demographic composition, controlling for age, education, urbanicity, and other covariates should eliminate the gaps. If local social conformity shapes behavior, we should observe individuals shifting toward the norms of their local community regardless of their own partisan identity—Democrats in Republican areas behaving more like Republicans, and vice versa. If cross-cutting social contexts reduce polarization, partisan gaps should be smaller when partisans live among members of the opposing party than when they live among co-partisans.

A key question is whether local context fully determines behavior or whether partisan identity exerts an independent effect. If context dominates, partisans in the same environment should behave similarly—a Democrat in a Republican-majority area should be nearly indis-

tinguishable from their Republican neighbors. If partisan identity dominates, gaps should persist even within shared contexts: Democrats and Republicans living in the same types of areas should still behave differently from one another, even if both groups shift their behavior in response to local conditions. The persistence of partisan gaps across multiple contextual dimensions—partisan composition, urbanicity, region, and policy environment—would provide strong evidence that partisan identity drives health behavior independent of local culture. It would also suggest that behavioral polarization is "portable": people carry their partisan patterns with them regardless of where they live, implying that geographic mixing alone will not reduce polarization.

An alternative explanation emphasizes information rather than identity. Republicans and Democrats may have been exposed to different information about the pandemic through divergent media ecosystems and elite messaging [CITE]. If so, partisan differences in behavior could simply reflect partisan differences in perceived severity: Republicans behaved less cautiously because they genuinely believed the pandemic was less dangerous, based on the information they received. This explanation generates specific predictions. First, we should observe partisan differences in perceptions of pandemic severity—which prior research confirms [CITE]. Second, and critically, perceptions should strongly predict behavior: those who perceived the pandemic as more severe should have behaved more cautiously. If information and perceptions fully mediate the relationship between partisanship and behavior, controlling for perceptions should substantially reduce partisan gaps. However, if partisan identity shapes behavior through channels beyond information processing—through social conformity, symbolic expression of group membership, or resistance to outgroup norms—behavioral gaps may persist even among partisans with similar perceptions. A third possibility is that stated perceptions are themselves expressions of partisan identity: partisans may report what their group is "supposed" to believe about pandemic severity, even if their actual behavior reveals different risk assessments. If Republicans report low concern but behave more cautiously than their stated perceptions would predict, this would suggest that expressed perceptions are partly performative—reflecting what one says as a group member rather than genuine beliefs that drive behavior. We examine whether perceptions and behaviors move together, and whether behavioral gaps remain after accounting for perceived

severity.

Finally, we examine whether the severity of the pandemic moderates partisan polarization. If polarization thrives when threats are abstract but narrows when consequences become concrete and personal, partisan gaps should be smaller during periods of high mortality and incidence, when the pandemic's toll was most visible and immediate. Conversely, gaps may widen during periods of lower severity, when there was more room for motivated reasoning and partisan interpretation of ambiguous risk information. Such a pattern would suggest that reality can partially override partisan identity when stakes are high enough—but that polarization may return when the immediate threat recedes.

We address two primary research questions. First, are partisan differences in health behaviors—social contacts, mask-wearing, and vaccine acceptance—robust to policy environment, geographic context, and demographic composition? Second, under what conditions are partisan gaps in health behavior largest versus smallest? We examine variation across local partisan context, pandemic severity, and geographic and policy environments to identify the factors that amplify or attenuate partisan polarization in health behavior.

Our findings reveal that partisan differences in health behaviors are remarkably robust. Gaps persist within policy environments, across regions, in both urban and rural areas, and regardless of local partisan composition. At the same time, the magnitude of polarization varies in ways that speak to the underlying mechanisms. We discuss the implications of these findings for understanding how social identity shapes health behavior and for designing public health communication strategies that can bridge partisan divides.

2 Data and Methods

Data Source

We use data from the Berkeley Interpersonal Contact Study (BICS), a nationally representative survey that collected detailed information on social contact patterns, protective behaviors, and demographic characteristics during the COVID-19 pandemic in the United

States. BICS recruited respondents from an online panel and used weighting methods to generate nationally representative estimates. Respondents were asked how many people they had face-to-face contact with on the day before the survey interview, and detailed information was collected on up to three of these contacts, including information about each contact's age, sex, race/ethnicity, mask usage, and physical distance. Additional details on the survey methodology, questionnaire design, sampling, and weighting are described in Feehan & Mahmud (2021).

Sample Characteristics

We analyze data from BICS waves 1 through 6, spanning April 2020 to May 2021. Wave 0 is omitted because it did not record political affiliation in a comparable way. Political party identification was assessed by the panel provider prior to the survey. Upon enrollment in the panel, all potential survey respondents were asked to select their partisan affiliation from standard categories (Democrat, Republican, Independent, or Prefer not to Say/Other). We exclude respondents in the "Prefer not to Say/Other" category to prevent potential misclassification bias.

Table 1: Sample Characteristics by Political Party

Characteristic	**Republican**	**Independent**	**Democrat**
N	5,517	4,527	8,189
Mean age (years)	50.0	46.4	46.3
Male (%)	51.5	50.1	46.3
White (%)	89.6	78.4	69.7
College graduate (%)	-68.0	-70.6	-62.9
Metropolitan (%)	82.5	85.4	87.8
Employed (%)	58.6	55.1	59.8
Household size	2.91	2.88	2.87

The three partisan groups differed on several demographic characteristics (Table 1). Repub-

licans were older on average and more likely to be male. The most striking demographic difference was in racial composition: 85% of Republicans identified as white, compared to 71% of Independents and 60% of Democrats. Democrats were more likely to live in metropolitan areas and to hold a college degree. These demographic differences are important to consider when interpreting partisan gaps in health behaviors, as some behavioral differences may be partially attributable to compositional differences between groups.

As visible in the sample sizes, our analytic sample of 18,233 respondents overrepresents Democrats (44.9%) relative to national estimates of party identification during the study period—30% Democrat, 29% Republican, and 39% Independent (Gallup, 2021)—while underrepresenting Independents (24.8% vs. 39%). Republicans are represented at roughly their national share (30.3% vs. 29%). Survey weights are applied throughout our analyses to improve representativeness. These weights adjust for age, gender, race/ethnicity, and education, and we additionally rake on party identification to match national targets. However, because partisanship is highly correlated with these demographic characteristics, the party-adjusted weights produce estimates that are nearly identical to the original demographic weights for within-party comparisons (see Section 7.8).

Measures

2.0.1 Health Behaviors

Social Contacts. Our primary contact measure is the total number of face-to-face contacts respondents reported for the day prior to the survey interview (`num_cc`). BICS defined a contact as any in-person interaction involving either physical touch or a conversation of at least a few words. We also examine non-household contacts (`num_cc_nonhh`) as a supplementary measure that isolates discretionary social exposure outside the home, where respondents have greater behavioral latitude.

Mask-Wearing. Mask usage is derived from the contact-level alter roster. BICS collected detailed information on up to three non-household contacts per respondent; for each of these contacts, respondents indicated which protective behaviors—if any—were used during

that interaction, including wearing a face mask. We aggregate these reports to construct a respondent-level measure equal to the percentage of recalled non-household contacts during which the respondent wore a mask (0–100 scale). Because this measure is based on a subset of up to three contacts rather than all contacts reported, it should be interpreted as reflecting mask behavior during a representative sample of a respondent's non-household interactions. This measure captures actual masking behavior during specific interactions rather than general self-reported attitudes toward masks, and is therefore less susceptible to social desirability bias on abstract attitudinal items.

Vaccination. Vaccine acceptance was measured in Wave 6 (May 2021) only. We define vaccine acceptance as a binary indicator equal to one if the respondent was already vaccinated or stated an intention to vaccinate. Because it was assessed in a single wave, vaccine acceptance is not examined in the main analyses; descriptive partisan differences are reported in Section 7.1.

2.0.2 Political Affiliation

We measure political affiliation using respondents' self-reported party identification. Respondents were classified as Democrats, Republicans, or Independents based on their responses.

2.0.3 Geographic Context

We link individual survey responses to geographic data at the ZIP code and congressional district levels. This allows us to characterize the local partisan environment and policy context in which respondents were living.

Partisan Composition. We characterize the local partisan environment using congressional district-level vote share from the 2020 U.S. House elections. Each respondent is linked to their congressional district via ZIP code, and we record the Democratic and Republican two-party vote share in that district. We use these vote shares continuously in some models and in quartile-based categories in others. We also construct a binary indicator (`In_Opposing_Party_CD`) equal to one if the respondent's stated party identification dif-

fers from the party of their congressional district's representative, allowing us to examine whether partisan behavioral gaps persist when Democrats and Republicans are embedded in opposing-party political environments.

Policy Environment. We characterize local policy context using county-level mask mandate status, derived from a county-level dataset of public health orders. Mandates are classified into four categories: strict (required in all or most public places), less strict (required in indoor commercial settings or for employees), unspecified or unclear, and no mandate. This classification allows us to test whether partisan differences in health behaviors persist within comparable policy environments, which would indicate that individual partisan identity rather than top-down policy drives behavior. Importantly, this measure captures whether a county had a mandate in place at all rather than the precise timing or duration of mandates; we are therefore unable to assess whether behavioral differences narrowed immediately following mandate adoption or eroded over time as mandates persisted.

2.0.4 Control Variables

We control for a range of demographic and socioeconomic characteristics that may confound the relationship between partisanship and health behaviors, including age, gender, race/ethnicity, education, income, and urban/rural residence.

Analytic Approach

2.0.5 Descriptive Analysis

We first present descriptive statistics on partisan differences in health behaviors across survey waves. We use survey weights to generate nationally representative estimates.

2.0.6 Regression Models

We estimate regression models to assess partisan differences in health behaviors while controlling for potential confounders. For continuous outcomes (e.g., number of contacts), we

use linear regression. For binary outcomes (e.g., mask-wearing), we use logistic regression.

To examine whether partisans differentially responded to changes in COVID-19 severity over time, we estimate models interacting political party with incidence rates:

$$Y_{ict} = \beta_1 \text{Party}_i + \beta_2 \log(\text{Incidence}_{c,t-1}) + \beta_3(\text{Party}_i \times \log(\text{Incidence}_{c,t-1})) + \gamma_c + \mathbf{X}_i'\delta + \epsilon_{ict}$$

where Y_{ict} is the outcome (daily contacts or mask usage) for individual i in county c at time t, Party_i indicates partisan affiliation (with Democrats as the reference category), $\log(\text{Incidence}_{c,t-1})$ is the log-transformed COVID-19 incidence rate in county c during the week *prior* to the survey, γ_c represents county fixed effects, and $\mathbf{X}_i$ includes individual-level controls (interview day, urban/rural residence). We use lagged incidence (and mortality) rates to ensure proper temporal ordering: respondents observe pandemic conditions in their county, and then adjust their behavior in the following week. The coefficient β_3 captures differential partisan responsiveness to these observed conditions.

To facilitate comparison between incidence and mortality effects, we standardize both predictors (z-score) before estimation. Coefficients thus represent the change in behavior associated with a one standard deviation increase in log incidence or log mortality.

The county fixed effects are central to our identification strategy. By including county fixed effects, the model uses only within-county variation over time, answering the question: "within the same county, as incidence rises and falls, how do partisans adjust their behavior?" This approach controls for all time-invariant county characteristics—local political culture, healthcare infrastructure, population density, and baseline behavioral norms—that might otherwise confound the relationship between incidence and behavior.

We intentionally omit wave fixed effects from these models. Including wave fixed effects would absorb the temporal variation in incidence rates that we aim to study. Following standard practice for modeling rates that span orders of magnitude, we log-transform incidence rates before analysis. Effect sizes are reported for a 10% increase in the log rate.

2.0.7 Geographic Heterogeneity

We examine whether partisan differences vary across geographic contexts by estimating models with interactions between individual partisanship and local partisan composition.

2.0.8 Temporal Dynamics

We assess how partisan differences evolved over time by estimating models separately for each survey wave and by examining interactions between partisanship and survey timing.

Analytical Framework

Our analyses are organized around the competing theoretical explanations introduced above. Each explains partisan behavioral gaps differently, and each generates testable predictions about where and when gaps should be largest or smallest.

Policy and demographics offer the most straightforward alternative explanations: if partisan gaps reflect where partisans live or who they are, they should shrink within shared policy environments and after demographic adjustment. We test this by comparing gaps within strict, lenient, and no-mandate counties and by estimating models that control for age, gender, race, education, and urbanicity.

Local context and social conformity predict that partisans should shift toward the behavioral norms of their communities regardless of their own identity. If context dominates, a Democrat in a heavily Republican area should behave nearly indistinguishably from their Republican neighbors. If identity dominates, gaps should persist within every contextual setting—across regions, urban and rural areas, and opposing-party congressional districts. The persistence of gaps across multiple contextual dimensions would constitute evidence for what we call "portable polarization": behavioral patterns that travel with individuals regardless of their environment.

Information and risk perception offer a within-person alternative: perhaps partisans behave differently only because they assess risk differently. We test this by examining whether

gaps close among partisans who report the same level of concern, and whether stated perceptions track behavior or diverge from it in ways consistent with expressive rather than genuine belief.

Objective pandemic severity offers a potential override: if concrete, undeniable consequences reduce the space for motivated reasoning, gaps should narrow during high-mortality periods. We test this using within-county variation in incidence and mortality rates over time, examining whether partisan behavioral convergence tracks objective conditions rather than stated concern.

3 Results

We structure our results around the competing theoretical explanations outlined in the introduction. We first establish that partisan differences in health behaviors exist and are substantial. We then test whether these differences can be explained by partisan gaps in risk perception: do partisans who express the same level of concern behave similarly? Next, we examine whether objective pandemic severity moderates polarization, testing whether mortality and incidence rates override partisan behavioral patterns. Finally, we assess whether these differences can be explained by geographic context, policy environments, or local partisan composition—or whether partisan differences in behavior persist even when you would expect them to dissipate.

3.1 Establishing Partisan Differences in Health Behaviors

We find substantial partisan differences across all three health behaviors examined: social contacts, mask-wearing, and vaccination. These differences were consistent across survey waves and persisted even after controlling for demographic characteristics including age, gender, race, education, and urban/rural residence. Republicans reported the highest levels of social contact, the lowest rates of mask usage, and the lowest vaccine acceptance rates, while Democrats exhibited the opposite pattern. Independents generally fell between the

two partisan groups, though often closer to Republicans on contact behavior and closer to Democrats on vaccine acceptance. Figure 2 provides a visual summary of these differences.

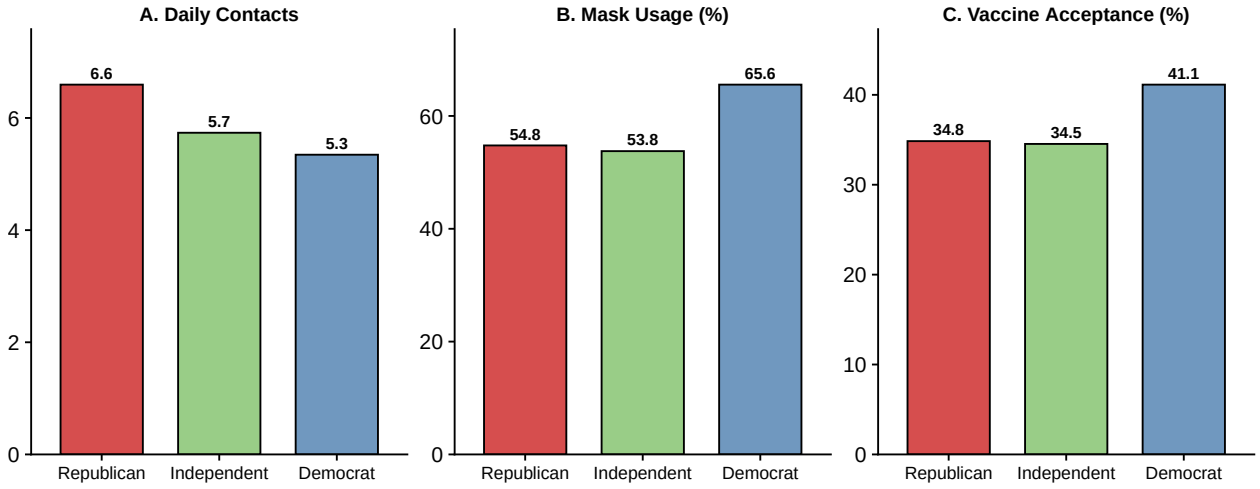

Figure 2: Partisan Differences in Health Behaviors. Panel A shows mean daily contacts, Panel B shows percentage of contacts during which masks were worn, and Panel C shows vaccine acceptance rates (Wave 6 only, May 2021). Vaccine acceptance includes those who were vaccinated or intended to get vaccinated. Values are weighted means using raked survey weights.

3.1.1 Social Contacts

Republicans reported significantly more daily interpersonal contacts than Democrats throughout the study period. On average, Republicans reported 6.6 total daily contacts compared to 5.3 for Democrats, a difference of approximately 1.2 contacts per day. This gap was even more pronounced for non-household contacts, with Republicans averaging 4.7 non-household contacts compared to 3.5 for Democrats. Independents fell between the two groups, reporting an average of 5.7 total daily contacts and 3.9 non-household contacts. These differences in contact patterns have important implications for disease transmission, as higher contact rates increase opportunities for viral spread within social networks.

3.1.2 Mask-Wearing

Partisan differences in mask-wearing were substantial and represented one of the largest behavioral gaps observed in the study. Democrats reported wearing masks during 65.6% of

their interpersonal contacts, compared to just 54.8% for Republicans—a gap of nearly 11 percentage points. Independents reported mask usage rates (53.8%) similar to Republicans. This partisan gap in mask-wearing was larger than differences observed across other demographic categories including race, gender, and education, highlighting the outsized role of political affiliation in shaping protective behavior during the pandemic.

Partisan differences in vaccine acceptance followed the same pattern (see Section 7.1); because vaccination was assessed only in Wave 6 and cannot be examined longitudinally or geographically with the same depth, the analyses that follow focus on contacts and mask-wearing. Underlying all three behavioral differences were substantial partisan gaps in pandemic risk perception (Figure 8 in Section 7.2): 63% of Democrats reported being "very concerned" about COVID-19 compared to 39% of Republicans—a 23 percentage point gap documented extensively in prior work (Gadarian et al., 2021; Kaushal et al., 2022; Milosh et al., 2021). The critical question is whether these perception differences can fully account for the behavioral gaps, or whether partisan identity shapes behavior through channels beyond information processing. We turn to that question next.

3.2 Does Perceived Pandemic Severity Mitigate Partisan Behavioral Polarization?

Democrats were substantially more likely than Republicans to report being "very concerned" about COVID-19 throughout the study period (Figure 8 in Section 7.2). Overall, 63% of Democrats expressed high concern, compared to just 39% of Republicans; Independents were closer to Republicans at 41%.

3.2.1 Stated Concern and Behavioral Gaps

If partisan differences in behavior are driven by differences in perceived pandemic severity, then partisans who express the same level of concern should behave similarly. Figure 3 tests this by comparing the Republican-Democrat gap in contacts and mask usage within each level of stated concern (see Figure 13 for levels by party).

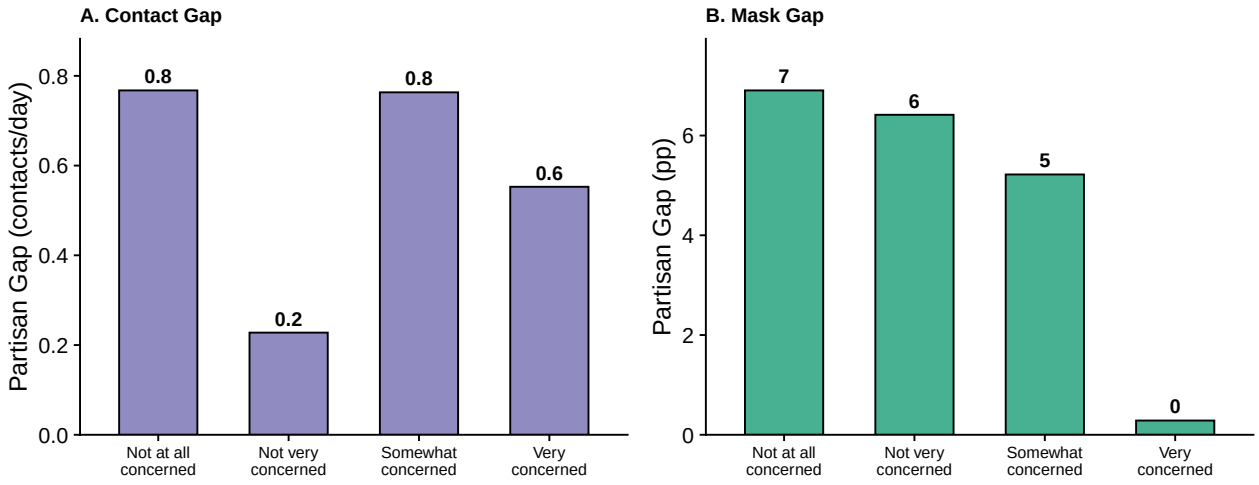

Figure 3: Partisan Gaps in Health Behaviors Within Each Level of Stated Concern. Panel A shows the absolute partisan gap in daily contacts; Panel B shows the absolute partisan gap in mask usage. For contacts, the partisan gap persists at all concern levels. For masks, the gap narrows substantially among the 'very concerned.' Sample sizes: Not at all concerned (n=1,127), Not very concerned (n=1,897), Somewhat concerned (n=5,894), Very concerned (n=9,168).

For social contacts, the partisan gap persists at every level of concern: Republicans report more contacts than Democrats whether they are "not at all concerned" or "very concerned" about the pandemic. Among the very concerned, Republicans still reported 5.6 daily contacts compared to 5.0 for Democrats, a gap of 0.6 contacts. For mask usage, however, perceived severity does appear to mitigate polarization: the partisan gap narrows as concern increases, and among the "very concerned," Republicans and Democrats report nearly identical mask usage (72% vs. 72%).

Interaction models formally test whether concern moderates the partisan gap (see Section 7.4 in the Appendix). For contacts, the Republican × Very Concerned interaction is statistically significant and negative (= -1.01, p = 0.03), indicating that concern does modestly narrow the partisan contact gap. For masks, the interaction is large and significant (= 12.2, p $<$ 0.001). The mechanism is clear: among the very concerned, Republicans wore masks during 72% of contacts, nearly identical to Democrats (72%). But this convergence reflects Republicans responding more sharply to concern: Republicans increased mask usage by 43 percentage points from "not at all concerned" to "very concerned," compared to 36 percentage points for Democrats. Higher perceived severity narrows the mask gap primarily

because Republicans increase their masking at a steeper rate. In sum, perceived severity mitigates partisan polarization for both behaviors, though the effect is substantially larger for mask usage than for social contacts.

3.2.2 How Polarization Evolved Over Time

The cross-sectional analysis above asks whether stated concern explains the partisan gap at a given moment — it does not. But a complementary question is whether *changes* in objective pandemic severity drove *changes* in the gap over time: did the gap narrow when conditions worsened, as threat override partisan identity, and widen when conditions improved? To examine this, we track how partisan behavioral gaps evolved across the six survey waves and connect those patterns to local mortality rates partisans would have observed in real time.

Figure 4 displays health behavior levels by party across survey waves (left column) and the corresponding partisan gaps (right column).

The contact patterns reveal notable variation over time (Figure 4, top row). In the previous section, we documented a substantial overall partisan gap in contacts. But this aggregate difference masks important temporal dynamics. Two periods showed relatively small contact gaps: early in the pandemic (April–June 2020, gaps of 0.4–0.7 contacts) and during the winter 2020–21 surge (December 2020–February 2021, gaps of 0.3–0.6 contacts). Between these periods, polarization spiked: the contact gap reached 3.5 contacts in September 2020, nearly ten times larger than during the winter surge. By May 2021, the gap had widened again to 1.6 contacts.

Mask usage tells a different story (Figure 4, bottom row). Rather than narrowing during high-severity periods, the partisan mask gap grew steadily wider over the course of the pandemic. The gap started at 6 percentage points in April 2020, dipped slightly in September, then climbed to 11 points by December and 15 points by February 2021, where it remained through May. Unlike contacts, mask polarization showed no convergence during the winter surge; if anything, it accelerated. This divergence suggests that mask-wearing became an increasingly entrenched marker of partisan identity over time.

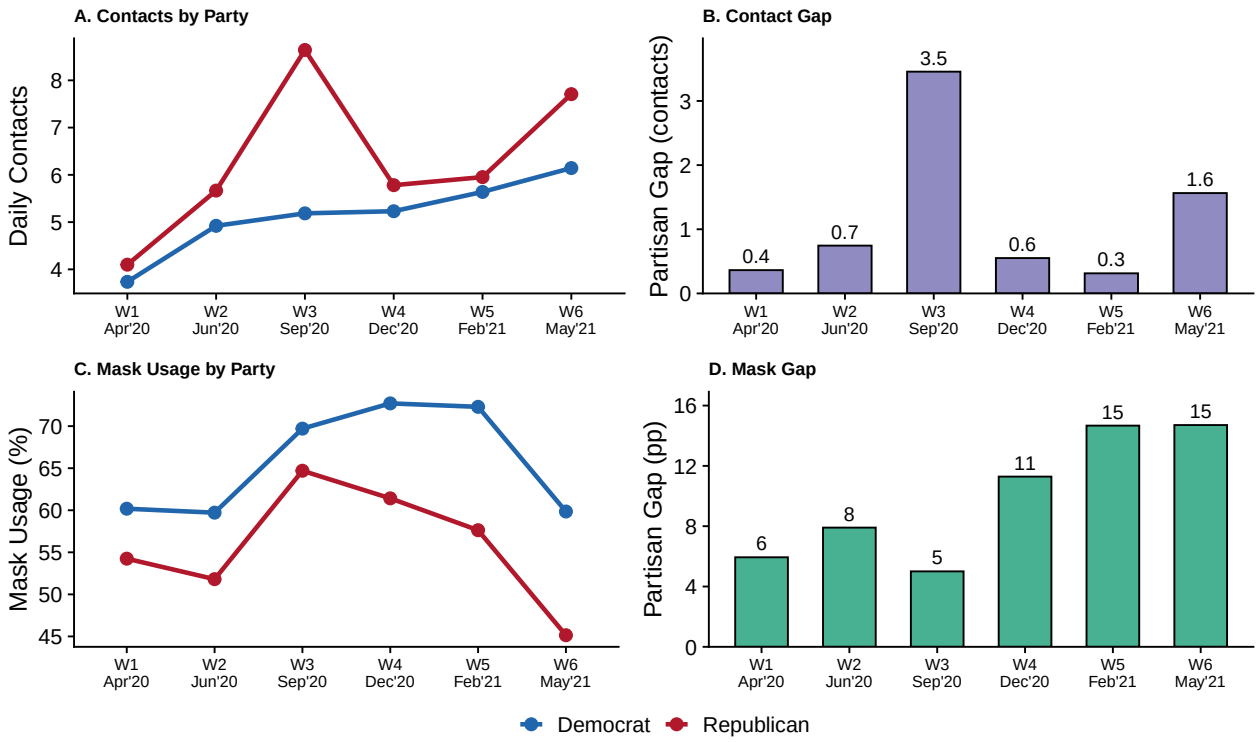

Figure 4: Health Behaviors and Partisan Gaps Over Time. Left column shows behavior levels by party; right column shows the absolute partisan gap. For contacts (top row), the gap widened through September 2020 before narrowing during the winter surge. For masks (bottom row), the gap grew steadily over time, suggesting increasing polarization.

What explains these divergent patterns? One possibility is that partisan gaps in behavior track pandemic conditions as they were reported at the time; specifically, mortality rates that partisans would have observed in real time.[1] The timing is suggestive: the periods of smallest contact gaps (early pandemic and winter surge) corresponded to periods of high or rising mortality, while the largest gap (September 2020) occurred when mortality was relatively low (Figure 5). This pattern suggests Republicans may be more responsive to changes in mortality rates, reducing contacts when deaths rise but also increasing contacts more quickly as rates decline.

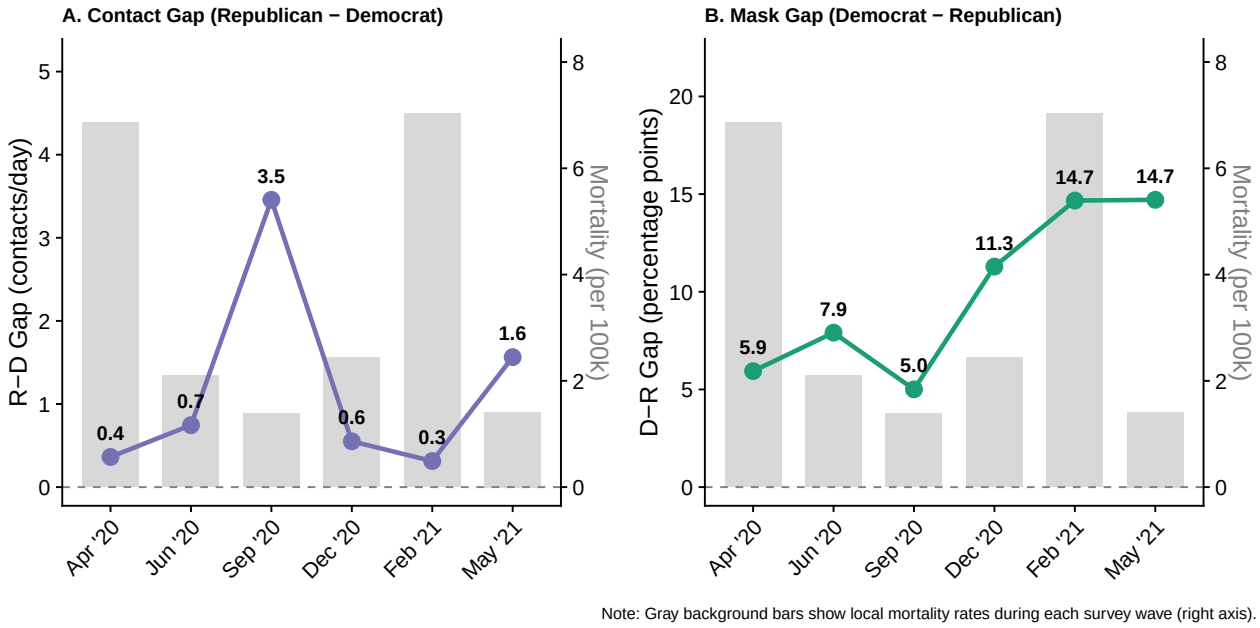

Figure 5: Partisan Gaps in Health Behaviors Across Survey Waves with Mortality Context. Lines show the Republican-Democrat difference in daily contacts (Panel A) and the Democrat-Republican difference in mask usage (Panel B). Gray bars in background show local mortality rates (deaths per 100,000) during each survey wave. The contact gap was smallest when mortality was highest (Feb '21), while the mask gap grew steadily regardless of mortality.

Regression models confirm this pattern. An interaction between mortality rates and party identification shows that Republicans are indeed more responsive to mortality than Democrats: a one standard deviation increase in local mortality is associated with a reduction of 0.33 contacts for Republicans, compared to 0.24 for Democrats (Figure 6; see

[1] We focus on mortality rather than incidence here because mortality showed stronger associations with contact behavior. See Section 7.11.1 for a detailed comparison of incidence and mortality effects.

Section 7.11 for detailed coefficients). Both groups reduce contacts when mortality rises, but Republicans adjust more sharply, consistent with the descriptive pattern of larger swings in Republican contact behavior across the pandemic.

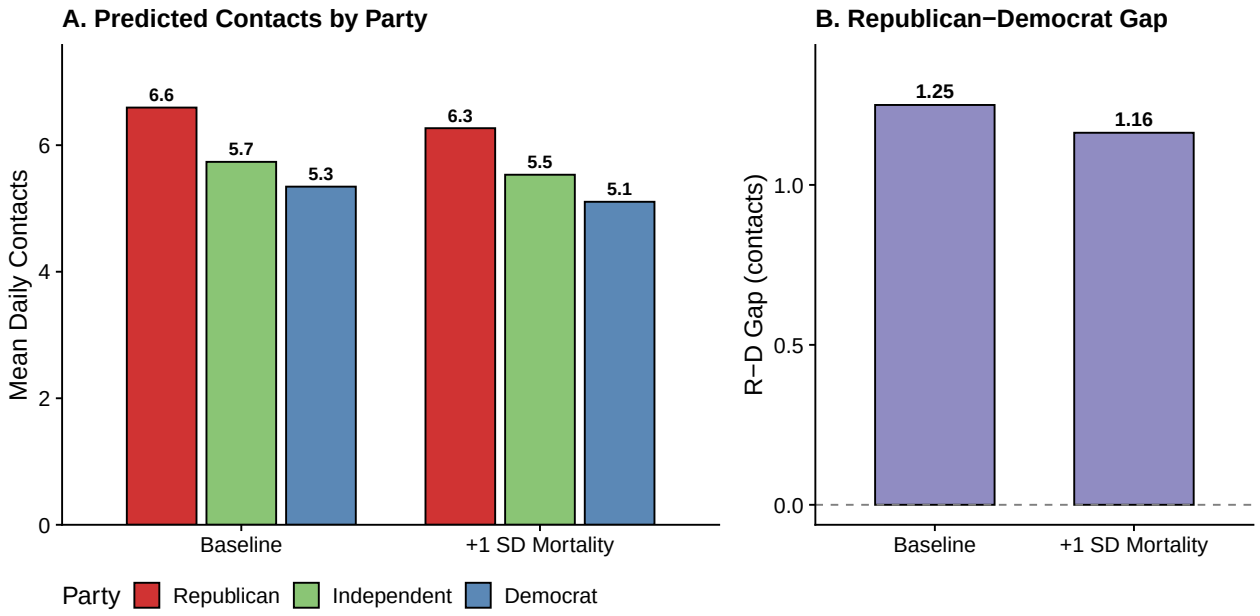

Figure 6: Contact Behavior Response to Mortality. Panel A shows predicted mean daily contacts by party at baseline (average mortality) and when mortality is one standard deviation higher. Panel B shows the Republican-Democrat gap in contacts at baseline and at higher mortality. All partisan groups reduce contacts when mortality rises, narrowing the partisan gap slightly.

3.2.3 When Attitudes and Behavior Diverge

Yet even as behavior responded to mortality, *stated perceptions* told a different story. Figure 12 in Section 7.2.1 displays each party's concern levels alongside the local mortality rates their respondents experienced. Early in the pandemic, perception and mortality moved together: in April 2020, when mortality was high, the partisan gap in concern was at its smallest, and Republicans were more similar to Democrats in their likelihood of reporting being "very concerned." But this alignment was short-lived. As the pandemic became politicized, stated concern decoupled from both mortality rates and behavior. The most striking evidence comes from February 2021: mortality reached its peak and the partisan gap in *behavior* narrowed to its smallest point in the study, yet Republican *stated concern* dropped

to its lowest (37%; see Figure 9 in Section 7.2). Republicans reduced their contacts when deaths surged, but simultaneously reported *less* worry about the pandemic. This disconnect suggests that stated perceptions became performative over time, reflecting what partisans felt they *should* say rather than the risk assessments actually guiding their behavior. Stated concern, in other words, increasingly functioned as a marker of partisan identity — a signal of group allegiance — even as underlying behavior continued to respond to objective threat. The implication is significant: cross-sectional analyses linking stated concern to behavior may partly capture partisan performance rather than genuine risk assessment, and partisan gaps in *attitudes* may overstate the true divergence in the beliefs actually guiding behavior. For additional analysis of behavior among the "very concerned" over time, including how the contact and mask gaps among this subgroup evolved differently, see Figure 14 in Section 7.2.2.

3.3 Does Local Context Mitigate Partisan Behavioral Polarization?

The preceding analysis focused on how pandemic severity, whether perceived or objective, shaped partisan behavioral differences over time. We now turn to whether *where* people live affects partisan polarization. If partisan gaps reflect local policy environments, regional culture, or adaptation to local norms, they should narrow or disappear within shared geographic contexts.

3.3.1 Geographic and Policy Variation

Figure 7 displays Republican-Democrat gaps in contacts and mask usage across census regions, adjusted for survey wave and local mortality rates (behavior levels by party and region are shown in Figure 34 in Section 7.12).

Partisan gaps persist across every region: Republicans reported more daily contacts and lower mask usage than Democrats regardless of where they lived. The contact gap was largest in the West (1.7 contacts) and smallest in the South (0.9 contacts); the mask gap

was similarly largest in the West (14.6 pp) and smallest in the South (8.3 pp). Notably, the smaller gap in the South is driven primarily by Democrats reporting more contacts (5.5) than Democrats elsewhere (4.9 on average) — consistent with Democrats in a heavily Republican region adjusting toward local norms — rather than by Southern Republicans becoming more cautious.[2]

These regional patterns are robust to controls for local context and demographic composition. Partisan gaps persisted within each policy environment (Figure 37 in Section 7.12.2) and remained stable after adjusting for urbanicity and demographics: contact gaps ranged from 0.7 contacts in the South to 1.5–1.7 contacts elsewhere with minimal change across model specifications; mask gaps shrank modestly but persisted everywhere, with the West retaining a 12-point gap in the fully adjusted model (see Figure 23 in Section 7.7).

3.3.2 Behavior in Opposing-Party Contexts

The most direct test of local context effects examines whether partisans adapt their behavior when living among members of the opposing party.

Partisans in opposing-party congressional districts do adjust their behavior toward local norms — Democrats in Republican-led districts reported 0.1 more daily contacts and 6 percentage points lower mask usage than Democrats in Democratic-led districts, with comparable shifts in the opposite direction for Republicans — but partisan gaps persisted in both contexts (see Figure 25 in Section 7.7.2 for full estimates). Republicans reported more contacts than Democrats in both Democratic-led districts (gap: 1.4 contacts) and Republican-led districts (gap: 0.8 contacts); Democrats wore masks more than Republicans in both contexts (gaps of 9 and 7 percentage points, respectively). Local partisan context shifted the *level* of behavior for both groups without closing the gap between them. Taken together, the geographic evidence points consistently toward partisan identity rather than context as the primary driver of behavioral polarization: gaps persist within regions, policy environments, and local partisan contexts, and where they narrow, it is because members

[2]See Section 7.12 for additional geographic breakdowns including vaccination, perception, and finer-grained division and state-level maps.

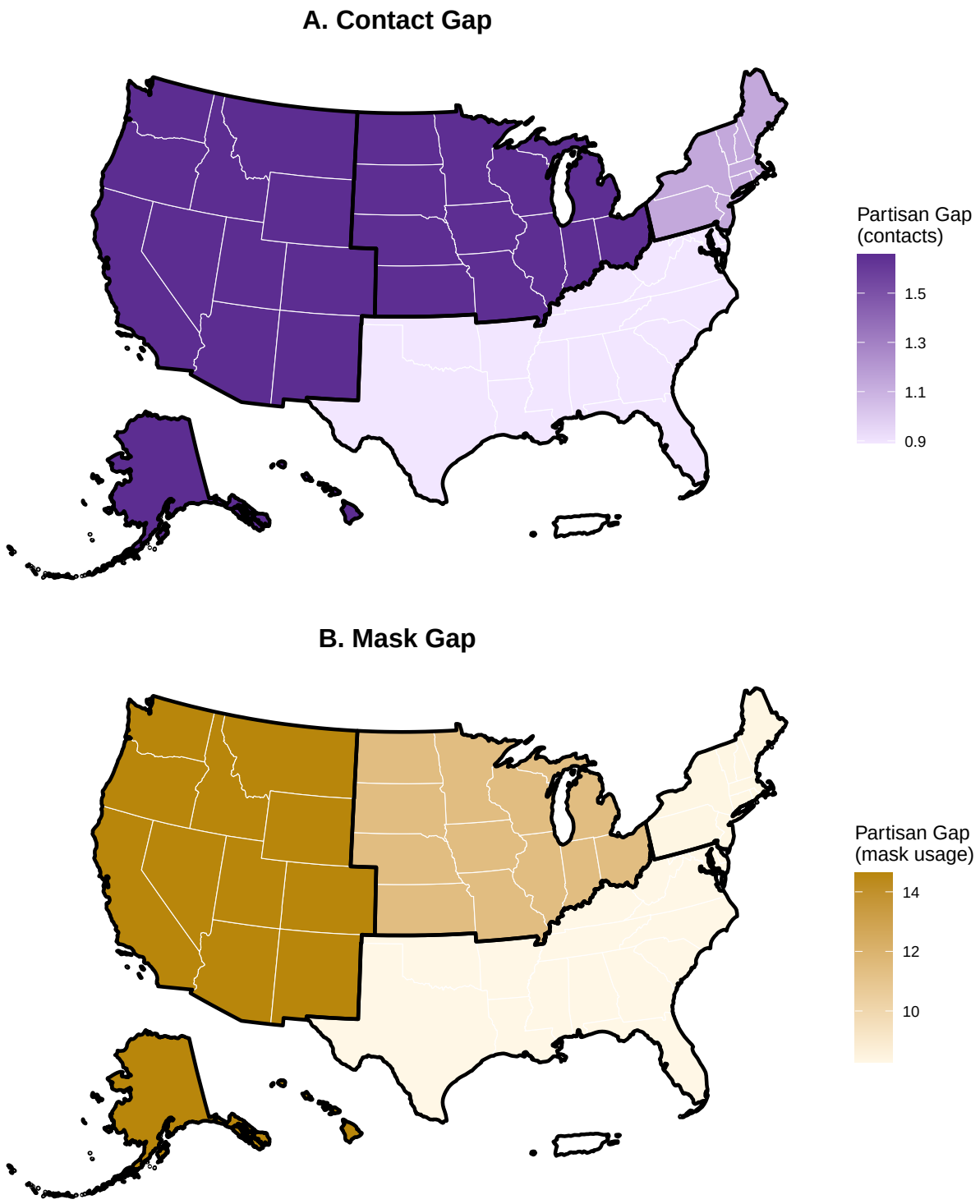

Figure 7: Geographic Distribution of Partisan Gaps by Census Region (Adjusted). Panel A shows the absolute partisan gap in daily contacts; Panel B shows the absolute partisan gap in mask usage. Estimates are adjusted for survey wave and local mortality rates. Darker shading indicates larger gaps. Black borders indicate census region boundaries.

of the minority party adjust toward local norms — not because the majority party becomes more cautious.

4 Discussion

4.1 Summary of Findings

Our findings reveal that partisan differences in health behaviors during the COVID-19 pandemic were not only substantial but often exceeded differences between other demographic groups. The Democrat-Republican gap in mask usage, for instance, was larger than differences by gender and, perhaps more surprisingly, larger than age-based differences, despite the elderly facing dramatically higher mortality risk. These partisan gaps cannot be dismissed as artifacts of where Democrats and Republicans happen to live: they persisted within the same policy environments, across different geographic regions, and regardless of local partisan context. Nor can they be explained by demographic composition; the gaps remained after adjusting for age, education, and other covariates. Republicans reported more daily contacts, wore masks less frequently, and expressed lower vaccine acceptance than Democrats across all survey waves, and these differences proved remarkably robust to every alternative explanation we tested.

Yet while partisan gaps proved robust to demographics, geography, and policy, they were not static over time. The magnitude of polarization shifted across different stages of the pandemic, responding to changing conditions and perceptions of severity. This temporal variation is theoretically revealing: if partisan differences were purely a matter of fixed group characteristics or stable ideological commitments, we would expect consistent gaps regardless of context. Instead, we find that pandemic severity, particularly local mortality, shaped the degree of partisan divergence, and it did so differently for different behaviors. Our regression analyses show that mortality rates were the strongest predictor of contact behavior for all partisan groups; incidence rates, by contrast, had no significant effect. This distinction matters: case counts are abstract statistics, easily dismissed or reinterpreted,

while deaths are concrete and visible, reported in local news, experienced through social networks, marked by obituaries and funerals. Partisan gaps in social contacts narrowed during high-mortality periods, particularly during the winter 2020-21 surge, suggesting that this kind of undeniable, proximate consequence can temporarily override partisan divergence. Mask-wearing, by contrast, showed *increasing* polarization over time, with the Democrat-Republican gap nearly tripling from 5 percentage points in September 2020 to 15 percentage points by May 2021. Unlike contacts, mask-wearing differences did not narrow when deaths surged.

What might explain these divergent patterns? The narrowing of contact gaps during high-mortality periods appears driven primarily by changes among those most concerned about COVID-19. When deaths surged, the threat became concrete and immediate, and respondents reporting high concern reduced their contacts sharply. Because Democrats were more likely to report high concern, this concern-driven response disproportionately affected Democratic behavior, pulling it closer to Republican levels during surges. The gap narrowed not because Republicans increased precautions, but because the most concerned (disproportionately Democrats) responded most strongly to rising mortality.

Wave 1 stands out as particularly distinctive. In April 2020, partisan gaps in both contacts and concern were at their smallest. This early convergence likely reflects multiple factors: the novelty and uncertainty of the initial outbreak, when the virus was new and partisan scripts had not yet solidified; but also the fact that the virus was especially deadly in these early months. Case fatality rates were at their highest before treatments improved and clinical protocols developed. The combination of novelty, uncertainty, and high lethality may have produced the conditions under which partisans responded similarly. As mortality rates declined and the pandemic became increasingly politicized, these gaps widened. The early pandemic also saw public health authorities shift their guidance on masks, initially advising against them before reversing course [CITE]. This reversal, combined with declining trust in public health institutions among both Democrats and Republicans over the course of the pandemic [CITE], may have contributed to the growing polarization. Yet shared erosion of trust cannot fully explain the partisan divergence: Republicans lost trust at substantially

higher rates than Democrats [CITE]. Importantly, this differential did not originate with COVID-19. Scholars have documented a growing partisan gap in trust toward scientific and public health institutions dating back to at least 2009 [CITE], a divide already evident in differential vaccine uptake during the H1N1 outbreak [CITE]. The pandemic, then, did not create partisan differences in trust; it activated and amplified a fissure that had been widening for over a decade.

Masks followed a different trajectory entirely. Unlike contact reduction, mask-wearing showed no narrowing during mortality surges. In fact, during the winter 2020-21 surge, when contact gaps reached their narrowest point, mask-wearing gaps were simultaneously approaching their peak. Partisan polarization, it seems, is not a fixed quantity that rises and falls uniformly, but a persistent force that finds expression where it can. This divergence may reflect the different nature of these behaviors. Reducing contacts is a relatively private decision with high personal costs: fewer contacts means fewer social interactions, less economic activity, and greater isolation. The costs of *not* reducing contacts during a surge are also personal and immediate. Mask-wearing, by contrast, is visible, low-cost, and had become deeply politicized by the time of the winter surge. The conditions that drove contact convergence simply did not apply.

Perhaps the most striking finding is the disconnect between stated concern and actual behavior. During the winter 2020-21 surge, Republicans reported their *lowest* levels of COVID-19 concern across all waves. Yet, this was when they reduced their contacts most substantially and where behavior converged the most. That is, expressed concern may have become partly performative: Republicans responding as they believed a Republican *should* respond, signaling appropriate group sentiment, while their behavior aligned with the objective risk environment. Notably, our survey never asked respondents about their partisan identity; this information was collected by the survey panel at a time before our survey was administered. The disconnect between what Republicans said and what they did was not an artifact of priming or survey context. This pattern is consistent with what we would expect if partisanship operates as a social identity: attitudes serve as markers of group loyalty, signaling allegiance to one's political tribe, while behavior responds to material consequences that

affect personal well-being.

Why, then, might masks have served as lower-stakes ground for partisan differentiation? Compared to contact reduction, wearing a mask is a low-cost action—it requires no fundamental change to one's social or economic life. But masks are also *visible*. Unlike the private decision of whether to attend a gathering, mask-wearing occurs in public spaces where others can observe it. This visibility transforms masks from a purely protective behavior into a potential signal of group membership. Wearing a mask—or conspicuously not wearing one—communicates something to others about who you are and what you believe, but mostly which group you belong to.

The scientific uncertainty surrounding mask efficacy may have amplified this dynamic. While the epidemiological evidence for reducing contacts was relatively straightforward—fewer interactions mean fewer transmission opportunities—the evidence on mask effectiveness was more contested, with guidance changing over the course of the pandemic and ongoing debates about mask types, fit, and real-world effectiveness [CITE]. This ambiguity created space for motivated reasoning: partisans could selectively attend to evidence that supported their group's position. When the science is clear and consequences are immediate, there is less room for partisan interpretation; when the science is contested and consequences are diffuse, identity can more easily shape both belief and behavior.

The result is that mask-wearing may have become a form of strategic interaction, a performance of partisan identity for an observing audience [CITE Goffman 1959]. If the goal is maintaining unity and standing within one's political tribe, then these performances are entirely rational, even if they elevate personal health risk. For Democrats, wearing a mask performed adherence to public health guidance, trust in scientific institutions, and solidarity with a collective response to the pandemic [CITE Dems trust science]. For Republicans, *not* wearing a mask could perform skepticism of government overreach, prioritization of personal freedom, and alignment with political leaders who downplayed the pandemic's severity [CITE Reps skepticism/freedom framing]. Crucially, these identity games were enabled by the relatively low stakes of mask-wearing: visible enough to signal tribal allegiance, yet low-cost enough that the personal consequences of "playing" remained modest. Contact

behavior offered no such opportunity. Reducing contacts is neither visible to one's coalition nor easily performed; it happens privately. And the stakes are higher, the consequences more concrete. This asymmetry could explain why mask usage became increasingly partisan over time regardless of changes in severity, while contact gaps responded to mortality: masks were terrain on which partisan games could be played; contacts were not.

The persistence of partisan gaps across diverse settings provides further evidence that identity, rather than context, drives these behaviors. If local environment were the primary determinant, we would expect Republicans in high-risk areas to dramatically reduce the gap between them and their Democratic neighbors; we would expect identical policy environments, such as mask mandates or stay-at-home orders, to eliminate behavioral differences; and we would expect distinct regional cultures, such as those in the South, to override partisan distinctions. Yet, partisan gaps persisted within high-mortality counties, within states with stringent public health measures, and across geographic regions with vastly different political cultures. This "portable polarization" suggests that partisan identity travels with individuals, shaping behavior regardless of where they live, what policies govern them, or what risks surround them. The gaps are not artifacts of Republicans and Democrats sorting into different environments; they are features of partisan identity itself.

Yet local context is not irrelevant. While partisan gaps persisted across settings, their magnitude varied in ways that suggest local environment still exerts some influence. In the South, for example, partisan polarization in health behaviors was smallest, not because Republicans became more cautious, but because Democrats in these regions behaved less cautiously, reaching toward Republican levels. This pattern raises a question: do partisans adjust their behavior to match co-partisans in their local context? The evidence suggests they do, and again, mask-wearing proves most susceptible to this adjustment. Democrats in Republican-leaning counties wore masks less frequently than Democrats elsewhere; Republicans in Democratic-leaning counties wore masks more frequently than Republicans elsewhere. Contact behavior showed weaker contextual effects. This pattern is consistent with our framework of strategic interaction: if belonging to one's local community carries value beyond national partisan allegiance, and if mask-wearing is the visible, low-stakes terrain on

which group membership is performed, then we would expect masks to be the domain where local context most shapes behavior. Partisans appear to balance two forms of belonging, adjusting their visible performances to fit both their partisan tribe and their geographic community.

Yet the story is not so clean. While Democrats become less cautious in Republican regions, so do Republicans. Local context does not simply shrink the partisan gap; it shifts the overall level of cautiousness for both groups. In Republican-leaning areas, Republicans may feel an even stronger pull to signal their partisan identity through reduced mask usage, while Democrats adjust toward local norms but maintain their relative position. The result is that the partisan gap persists even as the baseline shifts. This reinforces rather than undermines our central argument: partisan identity remains a crucial determinant of behavior even within local contexts that push everyone in the same direction. Local belonging and partisan belonging operate simultaneously, with local context setting the level and partisan identity preserving the difference.

These local context effects, however, are difficult to interpret causally. Two explanations are plausible. The first is selection: Democrats who happen to be more similar to Republicans in their attitudes and behaviors may move to, or choose to remain in, Republican-leaning areas for reasons that have little to do with partisan influence. The second is absorption: partisans living among opposite-party majorities may gradually absorb the values and behaviors of their local peers over time. Our cross-sectional data cannot distinguish between these mechanisms. Future research should address this question with panel data that follows the same individuals over time, allowing researchers to observe whether attitudes and behaviors shift as local contexts change, rather than relying on comparisons across individuals in different settings as we do here.

4.2 Implications for Public Health

Our findings show that partisan identity is a strong predictor not only of health behavior but also of stated concern about the pandemic. This pattern suggests that information asymmetry may have been at play. Through mechanisms of mass media, where partisan news

ecosystems deliver divergent messages about risk, or through network effects, where information flows more readily within than across partisan communities, Republicans may have received systematically different signals about pandemic severity than Democrats [CITE]. It is also possible that Republican-leaning regions are more geographically or socially clustered and isolated from other regions, creating structural barriers that prevent health information from disseminating easily into these communities.

These possibilities point to a broader research agenda. Understanding partisan networks along dimensions of partisan homophily, the tendency for partisans to associate with co-partisans, remains underdeveloped [CITE if available]. Equally important are the structural features of these networks: How clustered are partisan communities within themselves? How interconnected are they with ideologically dissimilar regions? Answering these questions has implications not only for designing more effective communication strategies to disseminate disease risk information across partisan divides, but also for improving disease transmission models. If partisan networks shape both information flow and physical contact patterns, then understanding their structure becomes essential for predicting how future outbreaks might spread through a polarized population.

These findings imply that effective public health policy aimed at reducing risk and encouraging protective behaviors should perhaps take partisan identity into account. Careful attention to how partisans differentially receive and respond to public health messages may be necessary for reaching the populations most resistant to adopting recommended behaviors. Previous research has explored tailoring health messages to specific demographic groups, such as by age, race, or education level [CITE]. Given our increasingly polarized society, it may be time to consider tailoring messages based on partisan identification as well, designing communications that resonate with the values, trusted messengers, and informational ecosystems of different partisan communities.

Our findings also suggest that perceived severity is a key driver of behavioral polarization. When Republicans perceived the threat as serious, as in the early weeks of the pandemic when lethality was highest, partisan gaps were smallest. Understanding the determinants of when Republicans consider a health threat more or less serious could translate into more

effective public health messaging. If communicators can identify frames, messengers, or types of evidence that increase perceived severity among Republicans without triggering partisan resistance, they may be able to narrow behavioral gaps. The challenge, of course, is that efforts to increase adherence among one group must not inadvertently decrease adherence among the other.

This tension points to a broader dilemma. By acknowledging partisan identity in public health messaging, we may make interventions more effective while simultaneously reifying the importance of partisanship in American society. Tailoring messages to Republicans and Democrats treats them as distinct groups with fundamentally different orientations toward health and science, potentially further stratifying society along partisan lines. Public health experts will need to decide: do they treat polarization itself as a public health problem to be addressed, working to bridge divides and rebuild shared trust in scientific institutions? Or do they acknowledge polarization as a fact of American life and operate within it, accepting partisan divisions as given in order to reduce the loss of life they produce? A potential middle path may exist: crafting messages that appeal to shared values and concerns, finding common ground that resonates with both Republicans and Democrats without alienating either. Such an approach would require identifying frames that transcend partisan identity rather than targeting it, emphasizing aspects of health threats that both groups find compelling. This is not merely a strategic question but an ethical one, with implications that extend beyond any single health crisis.

Any such effort must also consider the partisan composition of the scientific and public health fields themselves. Scientists are disproportionately likely to identify as Democrats, and public health professionals are even more so [CITE]. This homogeneity may contribute to the perception of science and public health as partisan institutions, a perception that either validates trust among Democrats or undermines it among Republicans. Part of the solution may involve actively seeking partisan diversity both among those who produce scientific knowledge and among those who disseminate it. A more diverse field may be less likely to be perceived as a partisan institution, and more likely to generate messages that resonate across partisan lines.

4.3 Limitations

Several limitations should be considered when interpreting our findings.

First, our data are observational and cross-sectional, which limits our ability to make causal claims. We observe associations between partisan identity and health behaviors, but we cannot definitively establish that partisanship *causes* these behavioral differences rather than correlating with unmeasured confounders. Although we control for a range of demographic and contextual factors, residual confounding remains possible.

Second, our survey waves capture only a portion of the pandemic, from April 2020 to May 2021. We do not observe behaviors before the pandemic began, which would provide a baseline for understanding how partisans differed in their pre-pandemic contact patterns and health behaviors. Nor do we observe the later stages of the pandemic, including the Delta and Omicron waves, when vaccination became more widespread and pandemic fatigue may have altered behavioral patterns. Our findings speak to a specific period of the pandemic and may not generalize to its full trajectory.

Third, all behavioral measures are self-reported. Respondents may have over- or under-reported their contact counts, mask usage, or vaccination intentions due to social desirability bias or recall error. If such biases differed systematically by partisanship, for example, if Republicans were less likely to over-report mask usage, this could affect our estimates of partisan gaps. However, the consistency of our findings across multiple behavioral measures and contextual comparisons provides some reassurance that the patterns we observe are not artifacts of reporting bias alone.

Fourth, our assessment of partisan vaccine gaps is limited to a single cross-section from May 2021, early in the U.S. rollout when eligibility tiers and supply constraints varied substantially across states and counties. Because access was uneven, the observed partisan gap in vaccine acceptance may partly reflect differential opportunity rather than differential willingness. More importantly, a single wave prevents us from tracking how vaccine polarization evolved over time. Evidence from other sources suggests the partisan vaccine gap widened considerably through late 2021 and into 2022; our data speak only to a specific moment

in that trajectory and should not be interpreted as characterizing the full arc of vaccine polarization.

Fifth, while we examine variation across multiple contextual dimensions, we cannot observe actual disease transmission. We measure behaviors that are relevant to transmission risk, but we cannot directly link these behaviors to infection outcomes. The relationship between the behavioral patterns we document and the partisan mortality disparities that motivate this study remains inferential rather than directly observed.

5 Conclusion

6 References

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7 Appendix

7.1 Vaccine Acceptance

Vaccine acceptance, measured during the final survey wave in May 2021, showed pronounced partisan differences. We define vaccine acceptance as being vaccinated or intending to get vaccinated, which captures underlying attitudes toward vaccination independent of differential access during the early rollout period. Among Democrats, 41.1% accepted vaccines, compared to 34.5% of Independents and 34.8% of Republicans—a 6 percentage point gap. Because vaccination was assessed in a single wave, it cannot be examined temporally or geographically with the same depth as contacts and mask-wearing; this finding should be interpreted as a cross-sectional snapshot from May 2021, early in the U.S. rollout when supply constraints and eligibility tiers varied substantially across states.

7.2 Pandemic Risk Perceptions Over Time

The main text documents substantial partisan differences in pandemic risk perception, with Democrats more likely than Republicans to report being "very concerned" about COVID-19. Figure 8 shows the overall distribution of concern by party, pooled across all waves. This section then examines how these perception gaps evolved over the course of the pandemic, paralleling our analysis of behavioral gaps over time.

Figure 9 displays the percentage of each partisan group reporting being "very concerned" about COVID-19 across survey waves. Figure 10 shows the Democrat-Republican gap in this measure over time, with weekly COVID-19 cases shown in the background for context.

The perception gap was smallest at the beginning of the pandemic: just 11.3 percentage points in Apr '20. By Dec '20, the gap had grown to 28.8 percentage points. In April 2020, Republicans were nearly as likely as Democrats to report being "very concerned" about COVID-19; by June 2020, that gap had more than doubled, and it remained elevated throughout the remainder of the study period.

Critically, while incidence was relatively low in April 2020, *mortality rates were at their*

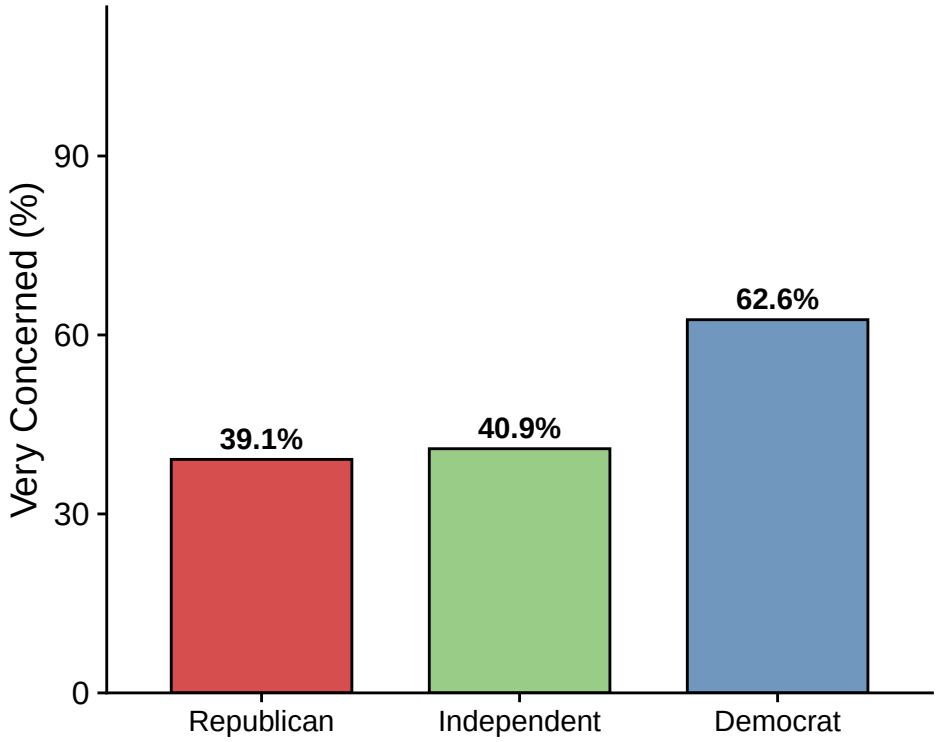


Figure 8: Partisan Differences in Pandemic Risk Perception. Bars show the percentage of respondents reporting being 'very concerned' about COVID-19, by political party. Values are weighted means using raked survey weights, pooled across all survey waves.

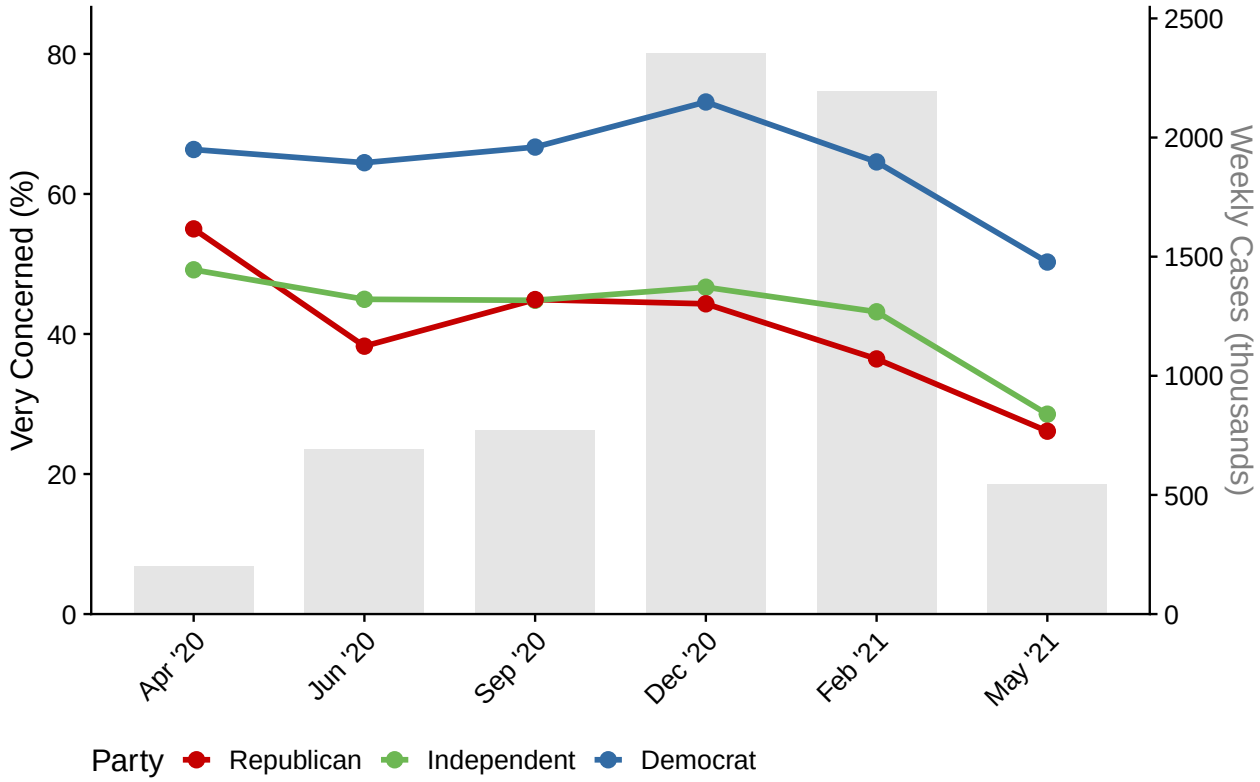


Figure 9: Pandemic Risk Perception by Party Across Survey Waves. Lines show the percentage of respondents reporting being 'very concerned' about COVID-19. Gray background bars indicate weekly COVID-19 cases (in thousands).

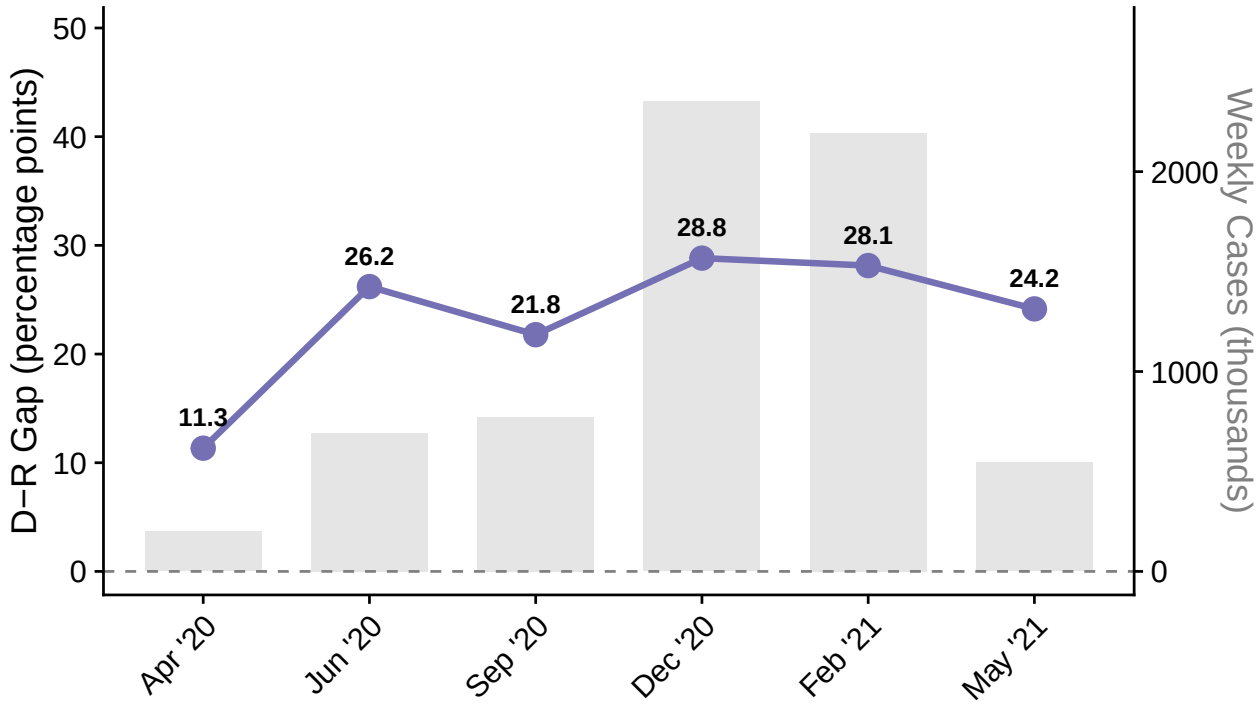


Figure 10: Partisan Gap in Pandemic Risk Perception Across Survey Waves. Points show the Democrat-Republican difference in the percentage reporting being 'very concerned' about COVID-19. Gray background bars indicate weekly COVID-19 cases (in thousands).

highest—the virus was simply more deadly in the early months before treatments improved and clinical experience accumulated. This pattern is consistent with our broader finding that concrete, visible consequences reduce polarization. When the threat was most lethal, perceptions were least polarized. As case fatality rates declined over subsequent months—even as case counts rose—the perception gap widened. The combination of declining lethality and escalating politicization of the pandemic through debates over lockdowns, masks, and the credibility of public health guidance likely both contributed to this divergence.

All three partisan groups showed declining concern over time, particularly from the winter surge through spring 2021 as vaccines became widely available. However, this decline was steeper among Republicans and Independents than among Democrats, maintaining the perception gap even as overall concern fell.

7.2.1 Perceptions vs. Local Pandemic Conditions

The previous figures show perceptions against *national* case counts. But Republicans and Democrats live in different places and thus experienced different local pandemic conditions. Figure 11 shows each party's stated concern alongside the average local incidence rate they actually experienced, given where respondents in each party lived. This allows us to assess whether perceptions tracked the objective conditions each group faced.

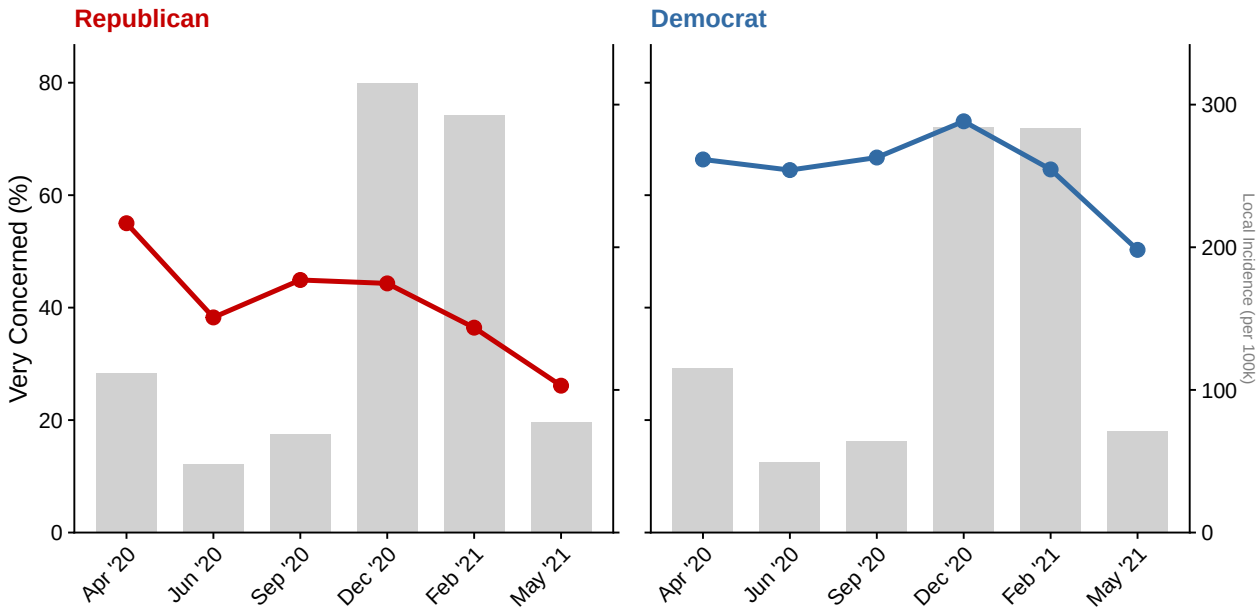


Figure 11: Perceptions vs. Local Pandemic Conditions by Party. Each panel shows one party's stated concern (line) against the average local incidence rate experienced by that party's respondents (bars). Local incidence is calculated as the weighted average of county-level weekly incidence rates among respondents of each party.

The figure reveals that Republicans and Democrats experienced similar local incidence trajectories—both groups saw the winter 2020–21 surge—yet their perceptions diverged over time. Notably, incidence was relatively low in April 2020 when the perception gap was smallest, complicating any simple story about incidence driving perceptions.

Figure 12 shows the same comparison using local mortality rates rather than incidence. Mortality is a more concrete and visible indicator of pandemic severity—deaths are reported in local news, experienced through social networks, and marked by obituaries—making it a stronger test of whether perceptions tracked objective conditions.

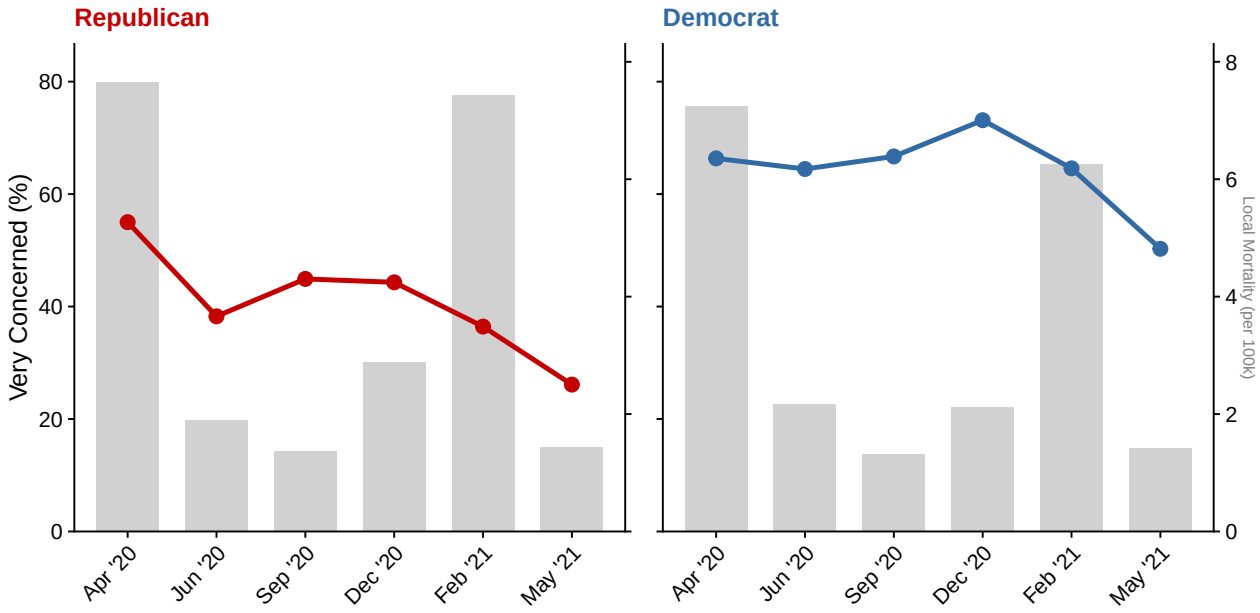


Figure 12: Perceptions vs. Local Mortality by Party. Each panel shows one party's stated concern (line) against the average local mortality rate experienced by that party's respondents (bars). Local mortality is calculated as the weighted average of county-level weekly mortality rates (per 100,000) among respondents of each party.

The mortality figure reveals a striking pattern: local mortality rates were actually *highest* in April 2020—precisely when the perception gap was smallest. The virus was simply more deadly in the early months before treatments improved and clinical protocols developed. As mortality rates declined over subsequent months, the perception gap widened. This pattern supports our broader thesis that concrete, visible consequences reduce polarization: when the threat was most lethal, partisans were most aligned in their concern.

The winter 2020–21 surge presents a more complex picture. Mortality rose sharply, and Republicans did reduce their contacts in response—yet their stated concern continued to decline. This disconnect between mortality-driven *behavior* and diverging *perceptions* suggests that by this point, expressed concern had become entangled with partisan identity in ways that were resistant to objective conditions. The early-pandemic alignment, when partisanship had not yet fully politicized COVID-19, may represent a window before symbolic boundaries hardened. Once the pandemic became a site of partisan conflict, stated perceptions became partly performative—reflecting group allegiance rather than risk assessment—even as underlying behavior continued to respond to concrete threat.

7.2.2 Health Behaviors by Concern Level

Figure 13 shows health behaviors by party within each level of stated concern. This complements Figure 3 in the main text by showing the absolute levels rather than the gaps.

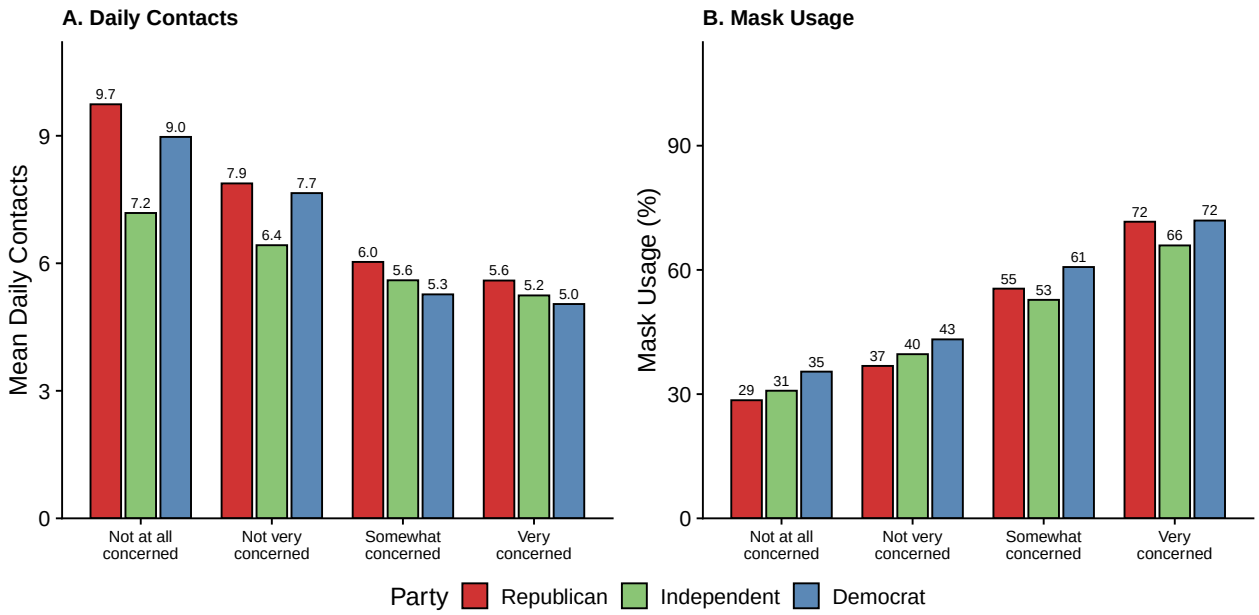


Figure 13: Health Behaviors by Party Within Each Level of Stated Concern. Panel A shows mean daily contacts; Panel B shows mask usage. Sample sizes: Not at all concerned (n=1,127), Not very concerned (n=1,897), Somewhat concerned (n=5,894), Very concerned (n=9,168).

Figure 14 shows how behavior among the "very concerned" subgroup evolved over time, alongside the changing share of each party reporting high concern. This analysis reveals that the meaning of "very concerned" changed over the pandemic as different types of partisans selected into this category.

Panel A shows that the share reporting "very concerned" declined for both parties, but especially for Republicans (55% to 26% vs. 67% to 49% for Democrats). Panels B and C reveal how behavior among the very concerned evolved—and the patterns differ in revealing ways.

For contacts (Panel B), very concerned Republicans initially reported *more* contacts than very concerned Democrats (gap of +1.4 to +3.4 contacts in waves 1-3). But this gap collapsed during the winter 2020-21 surge (wave 4-5) and actually reversed by waves 5-6, with

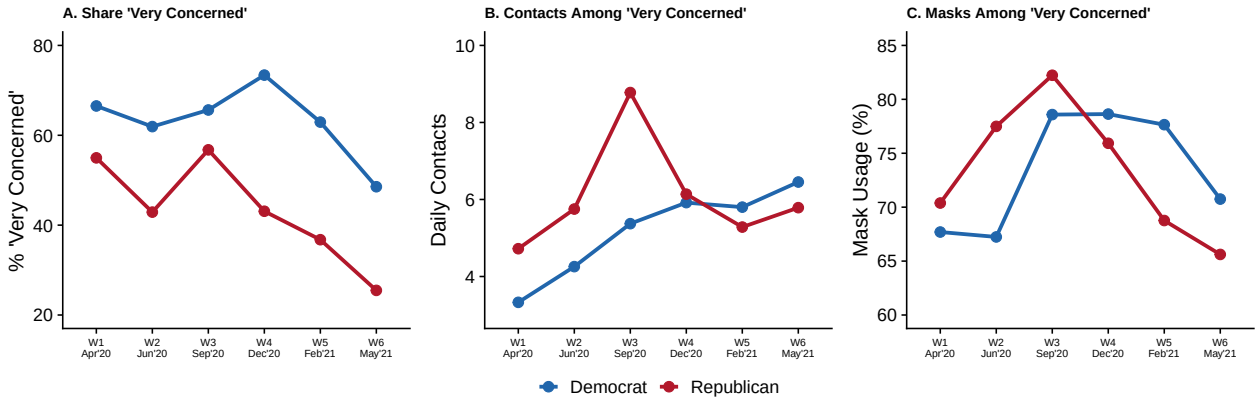


Figure 14: Behavior Among the 'Very Concerned' Over Time. Panel A shows the percentage reporting being 'very concerned'—declining for both parties but especially Republicans (55% to 26%). Panels B and C show behavior *only among those very concerned*. The contact gap (Panel B) was positive early on but collapsed during the winter surge and reversed by wave 5-6. The mask gap (Panel C) reversed earlier (wave 4) and persisted, suggesting politicization independent of concern level.

concerned Democrats reporting slightly more contacts. This timing aligns with the lifting of stay-in-place restrictions: among all respondents, the percentage reporting being under a stay-in-place order declined from approximately 95% in wave 1 to 70-81% by wave 5, with Republicans showing a larger decline than Democrats (Figure 17 in Section 7.3). The contact pattern suggests mortality-driven convergence early on, followed by Democrats increasing contacts as restrictions lifted—perhaps reflecting a strategic shift toward relying on masks and vaccines rather than contact reduction.

For masks (Panel C), a similar reversal occurred but with different timing: very concerned Republicans wore masks at *higher* rates than very concerned Democrats in waves 1-3, but by wave 4 Democrats wore masks more. Unlike contacts, this reversal did not correspond to the mortality surge or the lifting of restrictions—it persisted even as mortality declined in waves 5-6. This pattern suggests that mask-wearing became increasingly tied to partisan identity over time, such that even "very concerned" Republicans reduced masking as the behavior became symbolically marked.

7.2.3 Wave 1 Only: Gaps by Concern Level When Mortality Was Highest

The pooled analysis above combines data from all waves. To examine whether the relationship between concern and partisan gaps differed when objective threat was highest, Figure 15 replicates this analysis for Wave 1 (April 2020) only—the period when mortality rates were at their peak.

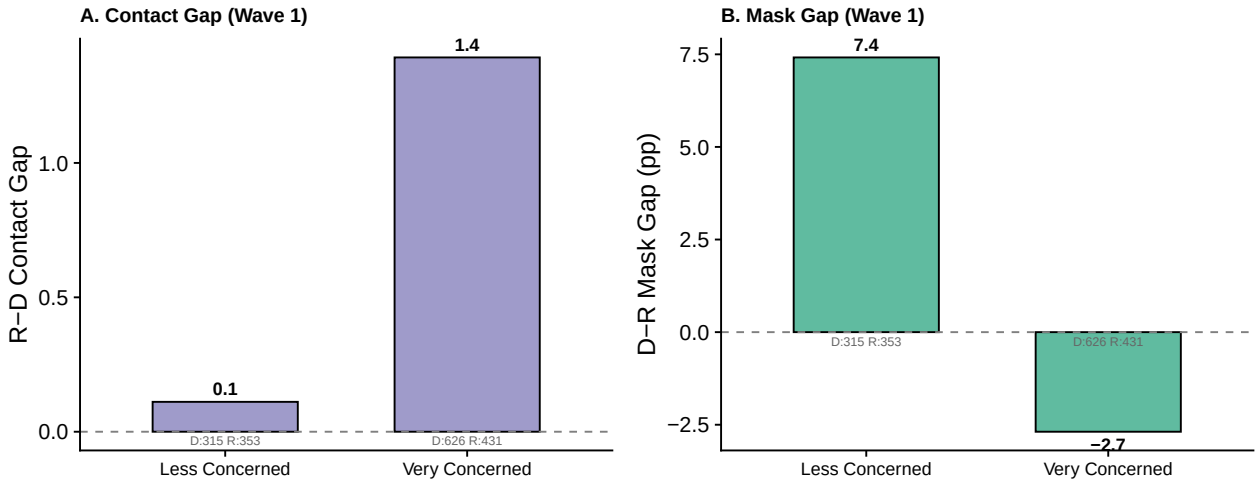


Figure 15: Partisan Gaps in Health Behaviors by Concern Level, Wave 1 Only (April 2020). Panel A shows the Republican-Democrat gap in daily contacts; Panel B shows the Democrat-Republican gap in mask usage. Wave 1 represents the period of highest mortality rates, before the pandemic became heavily politicized. Binary concern grouping (Very Concerned vs. Less Concerned) used for statistical power. Unweighted means used since survey weights were designed for the full sample. Sample sizes shown on each bar (D = Democrats, R = Republicans).

Comparing Wave 1 to the pooled results reveals whether the early pandemic—before extensive politicization—showed different patterns. If concrete mortality threat reduces polarization, we might expect smaller or even reversed gaps in Wave 1 compared to later periods.

7.2.4 The Politicization of Mask-Wearing: Exact Gap Values

Figure 14 above shows how mask-wearing patterns among the "very concerned" reversed over time, alongside the changing composition of who reports being very concerned. Figure 16 provides the exact D-R gap values for each wave.

As discussed in the main text, two dynamics likely contribute to this reversal: (1) masks

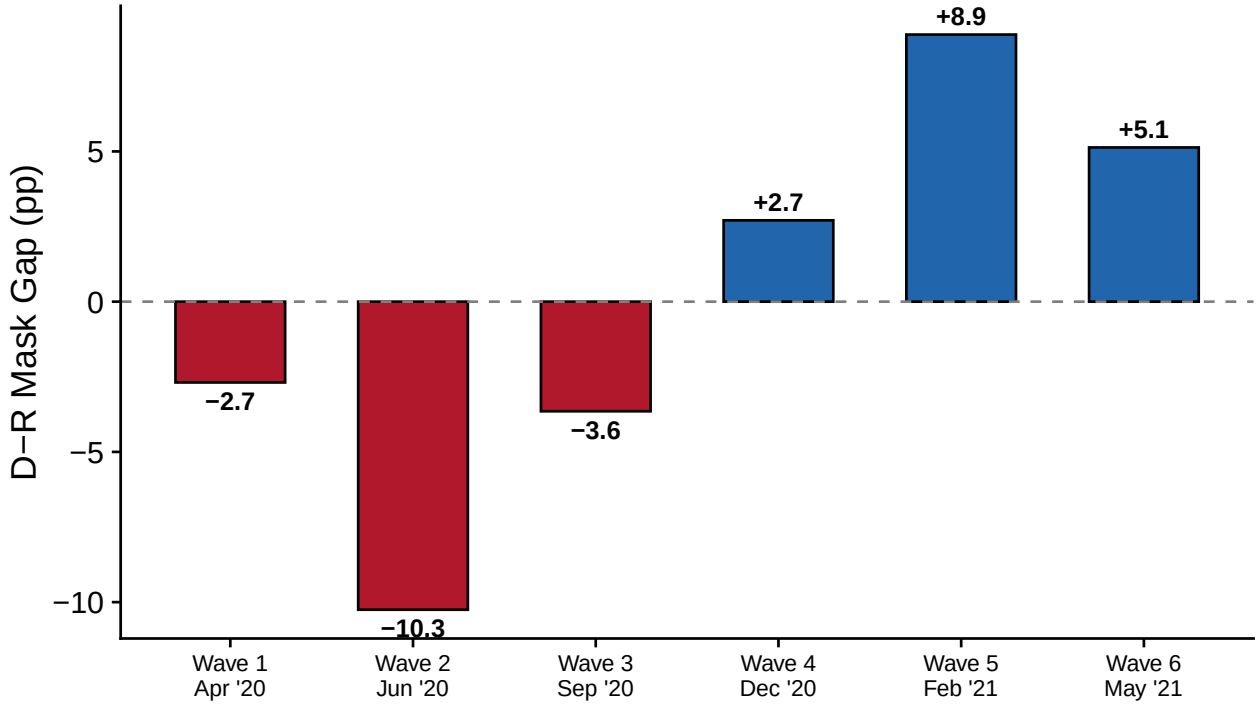


Figure 16: D-R Mask Gap Among 'Very Concerned' Respondents Over Time. Bars show the Democrat-Republican difference in mask usage among those reporting being 'very concerned' about COVID-19. Negative values (red) indicate Republicans wore masks more; positive values (blue) indicate Democrats wore masks more. The reversal from waves 1-3 to waves 4-6 is evident. See Figure 14 for context on changing composition.

became increasingly politicized as partisan signals, and (2) the Republicans who continued to report being "very concerned" became an increasingly select and unusual subset as overall Republican concern declined.

7.3 Stay-in-Place Orders Over Time

Figure 17 shows the percentage of respondents reporting being under a stay-in-place order across the study period. The BICS survey asked respondents whether they were currently under a stay-in-place order in waves 1-5, but dropped this question entirely in wave 6 (May 2021)—the only survey item discontinued during the study—reflecting the widespread lifting of restrictions by that time.

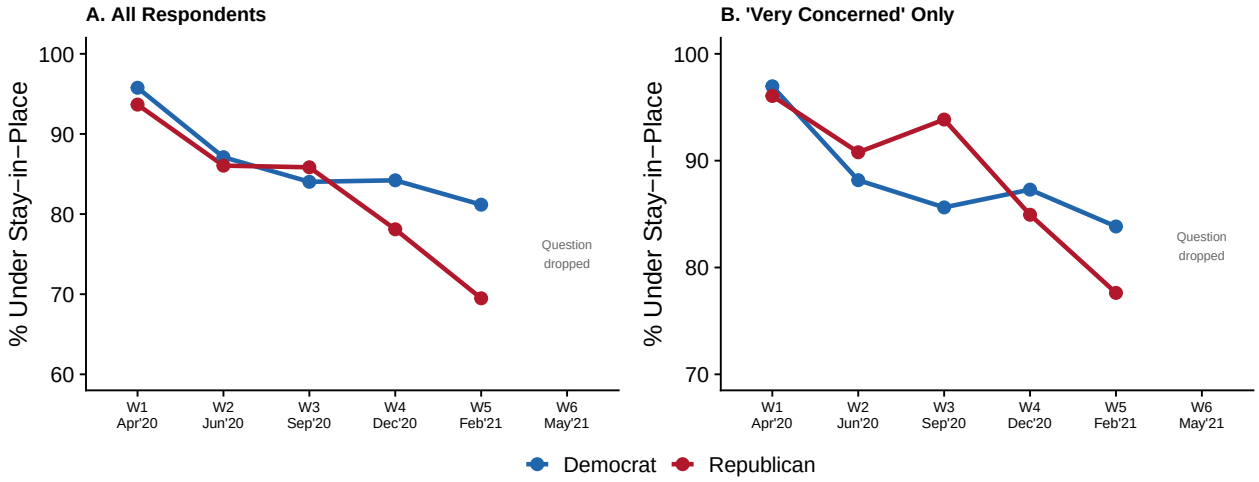


Figure 17: Percentage Reporting Being Under a Stay-in-Place Order Over Time. Panel A shows all respondents; Panel B shows only those reporting being 'very concerned' about COVID-19. Stay-in-place orders declined substantially over the study period. The question was dropped in wave 6 as most states had lifted formal restrictions by May 2021.

Among all respondents (Panel A), the percentage reporting being under a stay-in-place order declined from approximately 95% in wave 1 to 70-81% by wave 5, with Republicans showing a larger decline than Democrats (11 percentage point gap by wave 5). This overall pattern is more representative than the "very concerned" subset (Panel B), since the share of Republicans reporting being "very concerned" declined substantially over the study—from 55% in wave 1 to just 26% by wave 6—making the very concerned Republicans an increasingly selective group. The declining trend in stay-in-place orders provides context

for the behavioral changes observed in the main text: as restrictions lifted, particularly between waves 4-6, Democrats increased their contacts while maintaining high mask usage—consistent with a strategic shift toward relying on masks and vaccines for protection as they resumed social activities.

7.4 Interaction Models: Concern and Mortality as Moderators

To formally test whether stated concern and objective mortality rates moderate partisan gaps in health behaviors, we estimate interaction models of the form:

$$\text{Behavior} = \beta_0 + \beta_1(\text{Republican}) + \beta_2(\text{Concern}) + \beta_3(\text{Mortality}) + \beta_4(\text{Rep} \times \text{Concern}) + \beta_5(\text{Rep} \times \text{Mortality}) + \text{F}$$

In this specification, the partisan gap equals $\beta_1 + \beta_4(\text{Concern}) + \beta_5(\text{Mortality})$. A negative β_4 indicates that higher concern *narrows* the partisan gap; a negative β_5 indicates that higher mortality narrows it. Concern is measured as a binary indicator for reporting being "very concerned" about COVID-19; mortality is the standardized log of the previous week's county-level mortality rate.

We estimate two versions of this model to distinguish between national/temporal patterns and local variation:

1. **Wave FE model**: Includes wave fixed effects and controls for metropolitan status. This captures how the partisan gap varies with *national* pandemic severity (high-mortality waves vs. low-mortality waves).

2. **County FE model**: Includes county fixed effects. This asks: *within the same county*, when local mortality rises, does the partisan gap narrow? This isolates the effect of local conditions from all time-invariant county characteristics.

Figure 18 displays the results from both specifications.

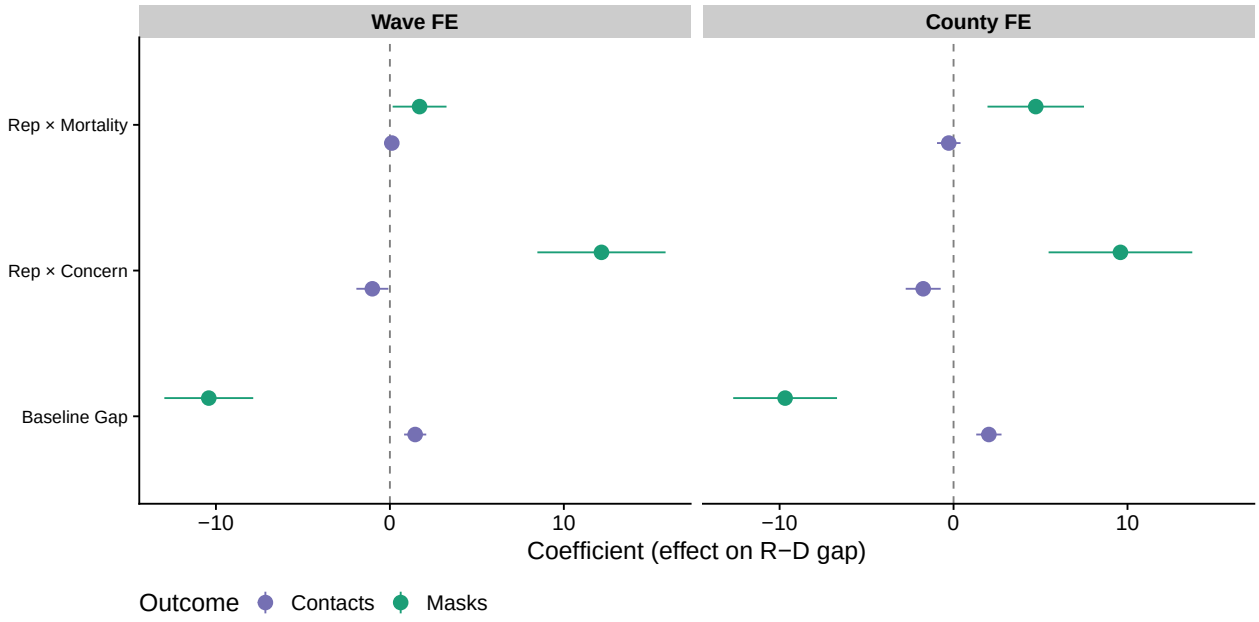


Figure 18: Moderators of Partisan Gaps: Wave FE vs. County FE Models. Points show regression coefficients from interaction models predicting contacts (purple) and mask usage (green). Left panel uses wave fixed effects (national/temporal variation); right panel uses county fixed effects (local variation). 'Baseline Gap' is the Republican main effect. 'Rep × Concern' shows how being very concerned changes the gap. 'Rep × Mortality' shows how a 1 SD increase in local log mortality changes the gap. Negative values indicate the gap narrows. Error bars are 95% CIs.

Wave FE Model (National/Temporal Patterns): For contacts, the baseline Republican-Democrat gap is 1.45 contacts per day. The concern interaction is statistically significant and negative (β = -1.01, p = 0.03), indicating that stated concern does modestly narrow the partisan contact gap. The mortality interaction is negative (β = 0.11, p = 0.573): during high-mortality waves, the partisan gap narrows.

For masks, the baseline gap is 10.4 percentage points. The concern interaction is large and significant (β = 12.2, p < 0.001): being "very concerned" substantially narrows the mask gap. The mortality interaction is smaller and in the opposite direction (β = 1.7).

County FE Model (Local Variation): The county FE model asks whether the partisan gap narrows *within counties* when local mortality rises. For contacts, the mortality interaction is -0.28 (p = 0.424), indicating that local mortality conditions also moderate the partisan gap. For masks, the local mortality interaction is 4.7 (p = 0.001).

Both specifications support the same conclusion: **concern moderates the mask gap but not the contact gap; mortality moderates the contact gap but not the mask gap.** This pattern holds whether we examine national trends (wave FE) or local variation (county FE), suggesting it reflects a genuine behavioral response to pandemic conditions rather than a compositional artifact.

7.5 Adjusted Partisan Behaviors Over Time

The main text presents descriptive partisan differences in health behaviors over time. To assess whether these patterns persist after accounting for demographic and contextual factors, we estimate regression models with party-by-wave interactions, controlling for age, race/ethnicity, Hispanic origin, gender, education, household size, congressional district partisan composition, and local mortality rates. We then extract estimated marginal means (emmeans) for each party at each wave, averaging over covariates. This approach isolates partisan differences while holding constant the demographic and geographic factors that might otherwise confound comparisons.

Figure 19 displays the adjusted behavior levels by party across survey waves, and Figure 20

shows the resulting partisan gaps.

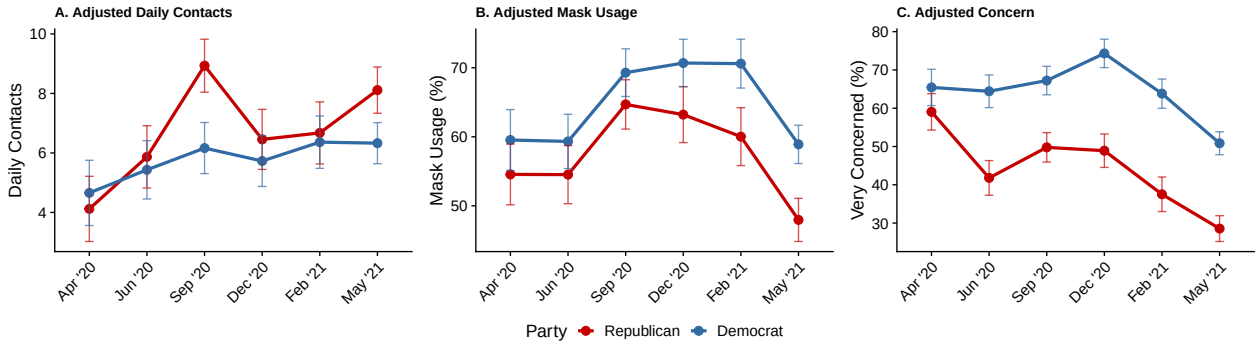


Figure 19: Adjusted Health Behaviors and Perceptions for Republicans and Democrats Across Survey Waves. Points show estimated marginal means from models controlling for age, race/ethnicity, gender, education, household size, congressional district partisanship, and local mortality rates. Error bars indicate 95% confidence intervals.

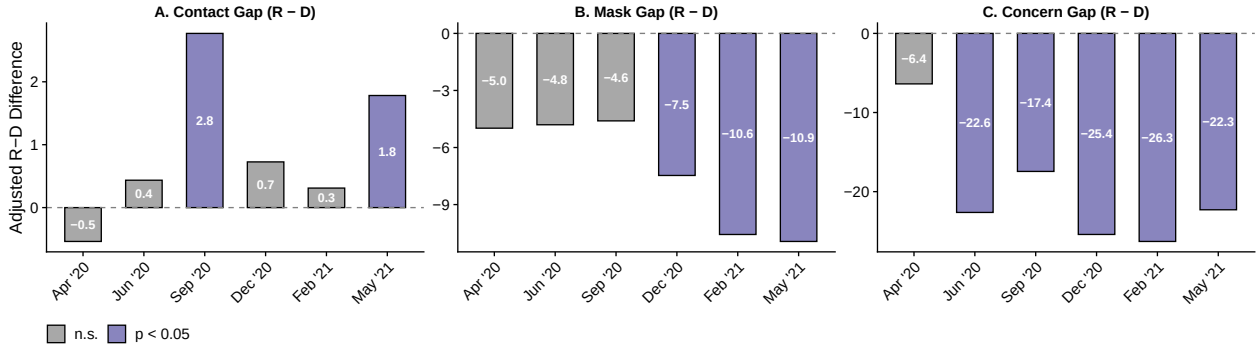


Figure 20: Adjusted Partisan Gaps in Health Behaviors and Perceptions Over Time. Bars show the Republican-Democrat difference in estimated marginal means for daily contacts (Panel A), mask usage (Panel B), and stated concern (Panel C), after controlling for covariates. Negative values indicate Democrats are higher. Purple bars indicate statistically significant gaps ($p < 0.05$); gray bars indicate non-significant gaps.

7.5.1 Interpretation

After adjusting for covariates, the patterns of partisan differences reveal important nuances. For **contact behavior**, the descriptive data show that partisan gaps were smallest early in the pandemic, with Republicans reporting only 0.4 more contacts than Democrats in April 2020 compared to gaps of 1-3 contacts in later waves. The adjusted analysis goes further: after controlling for demographics, the April 2020 gap is not only small but *statistically indistinguishable from zero* (see Figure 20, gray bars indicate non-significant differences).

In fact, the point estimate is slightly negative, suggesting that if anything, Democrats had marginally more contacts than Republicans when mortality was at its peak. This stands in stark contrast to the large, statistically significant gaps observed in September 2020 and May 2021, periods of relatively lower pandemic intensity. The pattern supports the interpretation that concrete, visible threat can eliminate—not merely reduce—partisan divergence in behavior.

For **mask-wearing**, the adjusted analysis reveals that partisan differences fluctuated rather than monotonically increasing. The Democrat-Republican mask gap (with Democrats wearing masks more) ranged from 4.6 percentage points in Sep '20 to 10.9 percentage points in May '21. Notably, the gap was relatively narrow during fall 2020 when mask usage was high across all groups, but widened again in early 2021 as vaccines became available and Republicans began to decrease mask usage more rapidly.

For **stated concern**, the adjusted analysis confirms that the perception gap was smallest at the beginning of the pandemic (when mortality was highest) and widened substantially thereafter. The adjusted Democrat-Republican concern gap ranged from 6.4 percentage points in Apr '20 to 26.3 percentage points in Feb '21. This pattern parallels the contact findings: when objective threat was most severe, partisan gaps in both behavior and perception were smallest. The subsequent widening of the concern gap, even as mortality rates declined, supports the interpretation that politicization drove perceptual divergence.

These adjusted estimates confirm that partisan differences in health behaviors and perceptions were not simply artifacts of demographic composition or geographic sorting. Even after controlling for a comprehensive set of covariates, substantial partisan gaps persisted throughout the pandemic, though their magnitude varied with pandemic conditions—narrowing when mortality was highest and widening as the pandemic became increasingly politicized.

7.6 Partisan Differences Within Policy Environments

A potential alternative explanation for partisan differences in health behaviors is that Democrats disproportionately reside in jurisdictions with stricter COVID-19 policies,

and their behavior simply reflects compliance with local regulations rather than partisan preferences. To address this concern, we examine whether partisan differences persist within the same policy environment.

Figure 21 and Figure 22 display mean contact rates and mask usage by party, stratified by county-level mask mandate policy (No Mandate, Less Strict, or Strict). If policy environment fully explained partisan differences, we would expect the gaps between parties to disappear within each policy category.

The results clearly demonstrate that partisan differences are not simply artifacts of differential policy exposure. Within each policy environment, Republicans reported more daily contacts than Democrats: the gap was 1.4 contacts in No Mandate areas, 1.0 contacts in Less Strict areas, and 1.3 contacts in Strict mandate areas. Similarly, Democrats reported higher mask usage than Republicans across all policy contexts: the gap was 9 percentage points in No Mandate areas, 11 percentage points in Less Strict areas, and 10 percentage points in Strict mandate areas.

These findings suggest that partisan identity shapes health behaviors above and beyond the effects of local policy. While stricter policies are associated with higher mask usage overall, the Republican-Democrat gap in mask-wearing remains remarkably stable across policy environments. This pattern is consistent with partisanship operating through mechanisms other than policy compliance—such as differential risk perception, social norms within partisan networks, or ideological commitments—that persist regardless of the local regulatory context.

7.7 Robustness to Context and Demographic Controls

This section presents a formal test of whether partisan gaps in health behaviors can be explained by local context (metropolitan status, county mask mandates) or demographic composition (sex, race, age, education). We estimate partisan gaps by census region using three nested models and examine whether gaps shrink toward zero as controls are added.

The results show that partisan gaps in both contacts and mask usage remain substantial

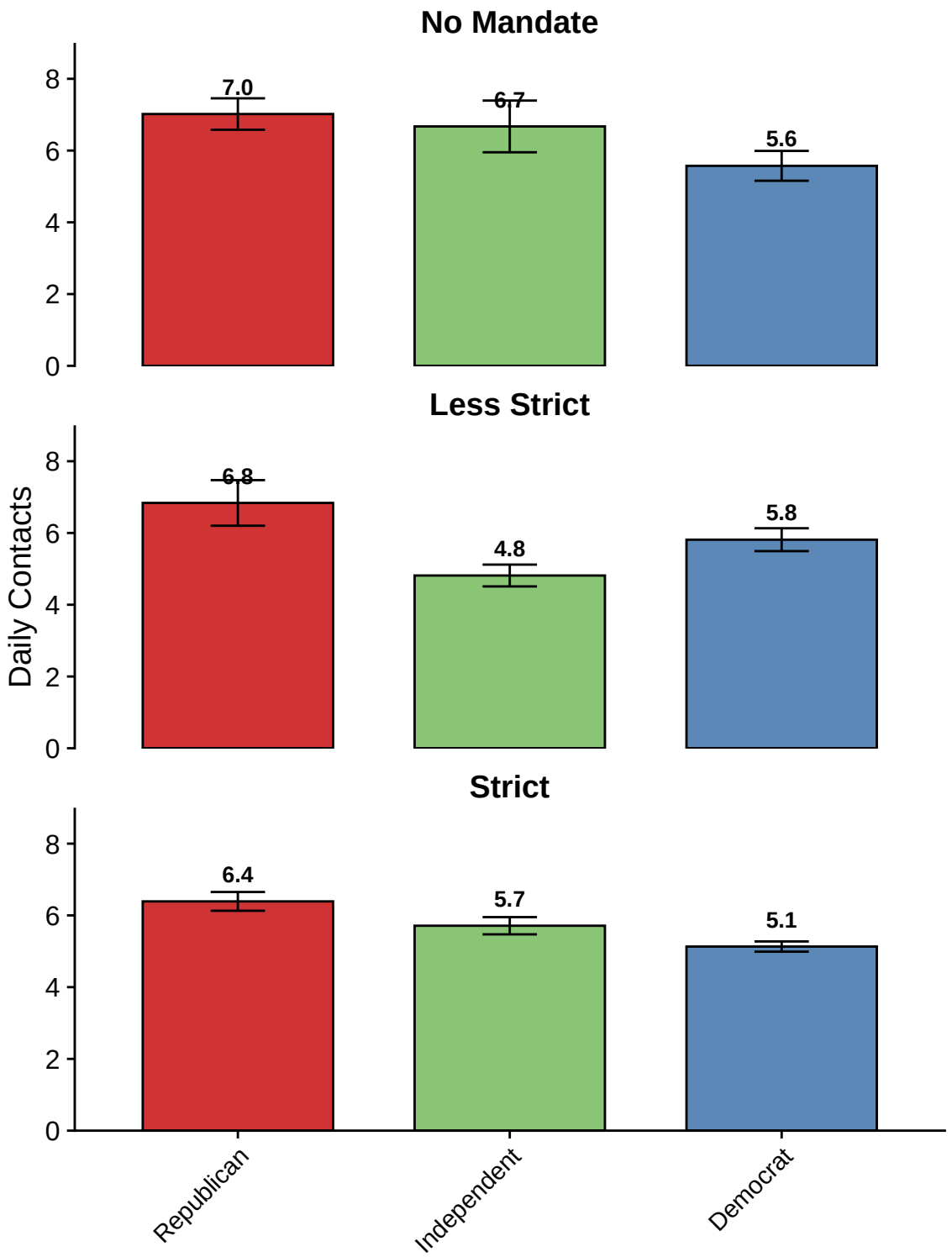


Figure 21: Daily Contacts by Party Within Policy Environments. Bars show weighted mean daily contacts for each party, faceted by county mask mandate policy. Error bars indicate standard errors.

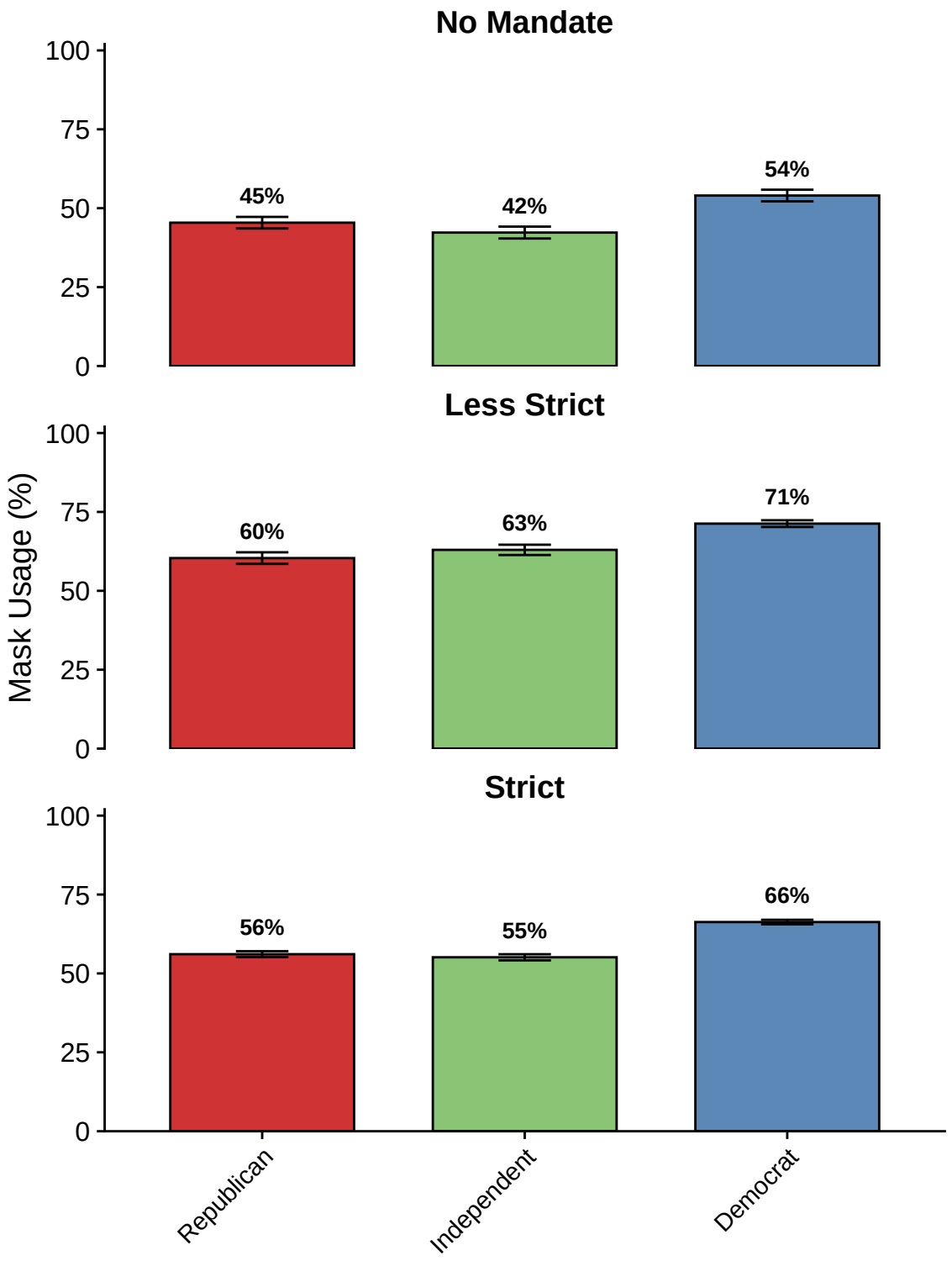


Figure 22: Mask Usage by Party Within Policy Environments. Bars show weighted mean mask usage (percentage of contacts with masks) for each party, faceted by county mask mandate policy. Error bars indicate standard errors.

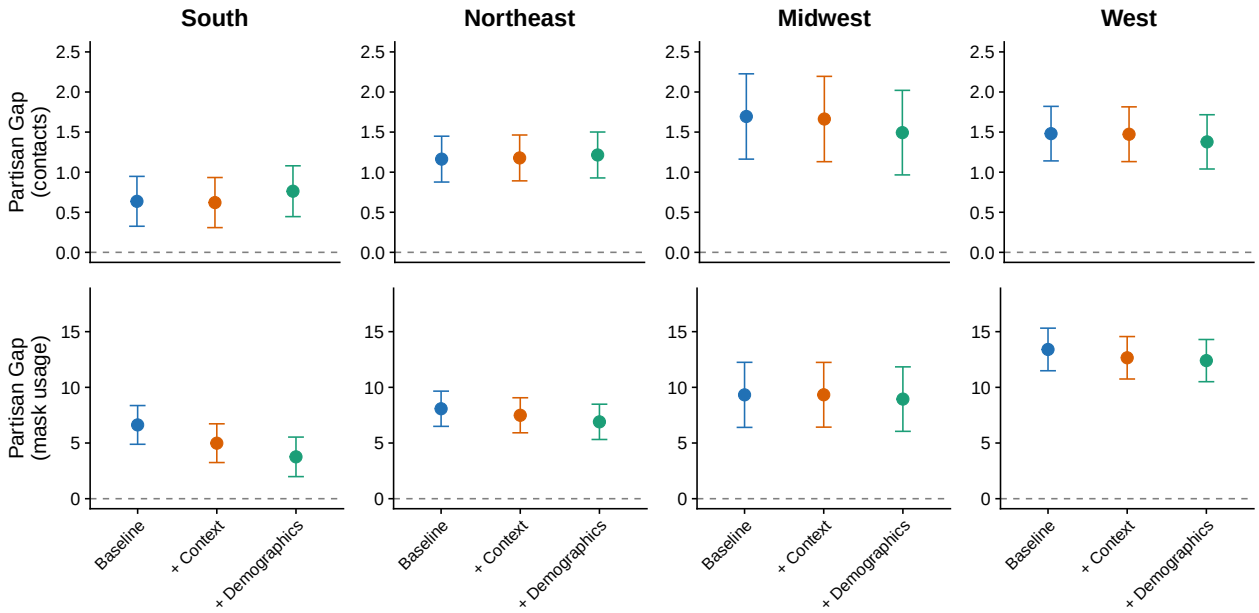


Figure 23: Effect of Controlling for Local Context and Demographics on Regional Partisan Gaps. Top row shows the absolute partisan gap in daily contacts (N = 12,753); bottom row shows the absolute partisan gap in mask usage (N = 9,087). Sample restricted to Democrats and Republicans; mask usage sample is smaller because it is conditional on reporting at least one contact with mask information. Each panel represents a census region. Points show gaps from three nested models: Baseline (wave + mortality), + Context (adds metro status and county mask mandates), and + Demographics (adds sex, race, age, and education). If context or demographics explain partisan gaps, the gaps should shrink toward zero as controls are added.

even after controlling for local context and demographics. For contacts, the gaps are remarkably stable across specifications, suggesting that neither urban-rural differences, policy environments, nor demographic composition can explain partisan behavioral differences. For masks, gaps shrink modestly—particularly in the South—but persist in all regions even in the fully adjusted model, with gaps ranging from approximately 4 percentage points (South) to 12 percentage points (West).

7.7.1 Partisan Gaps by Congressional District Context

Figure 24 presents a complementary view of the main text analysis (Figure 25), showing the absolute Republican-Democrat gaps in contacts and mask usage by congressional district partisan context across the three model specifications.

The gaps remain stable across model specifications in both district contexts. In Democratic-led districts, the contact gap ranges from 1.4-1.6 contacts across specifications; in Republican-led districts, it ranges from 0.5-0.7 contacts. For masks, gaps in Democratic districts range from 7-8 percentage points, while gaps in Republican districts range from 6-8 percentage points. These results complement the main text finding that partisan gaps persist regardless of local partisan context.

7.7.2 Robustness to Model Specification

Figure 25 displays predicted health behaviors by party and congressional district context, showing that partisan gaps persist in both Democratic-led and Republican-led districts.

Figure 26 demonstrates that partisan gaps in health behaviors persist regardless of whether we control for local context and demographic characteristics. The figure compares unadjusted estimates (wave only) with adjusted estimates (controlling for incidence, mask mandates, metro status, and demographics).

The adjustment for context and demographics has minimal impact on partisan gaps. For contacts, the Republican-Democrat gap shifts only slightly when adding controls: in Democratic districts, the gap remains approximately 1.3-1.5 contacts whether unadjusted or adjusted;

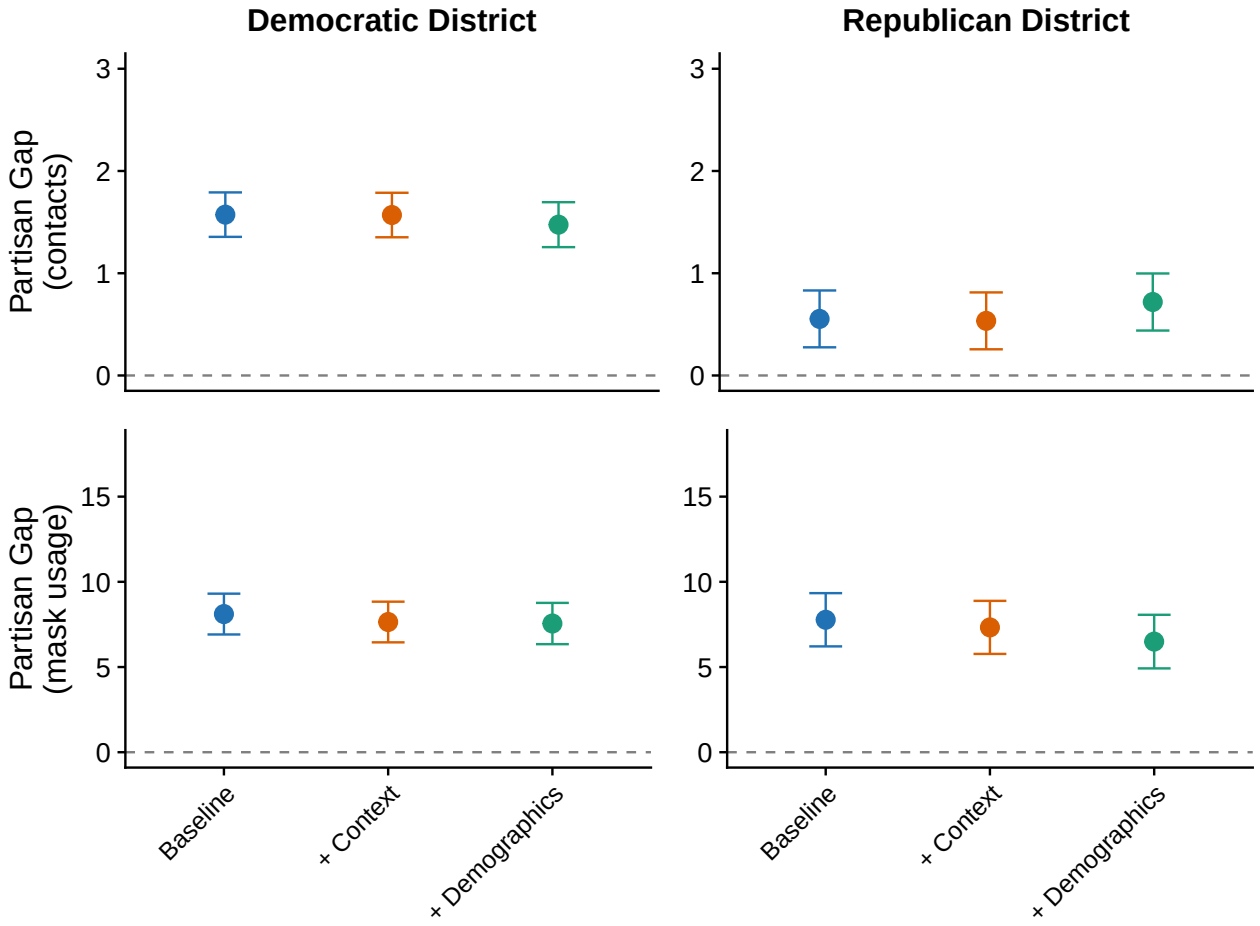


Figure 24: Partisan Gaps by Local Partisan Context Across Model Specifications. Left column shows gaps in Democratic-led congressional districts; right column shows gaps in Republican-led districts. Top row shows the absolute partisan gap in daily contacts; bottom row shows the absolute partisan gap in mask usage. Points show gaps from three nested models: Baseline (wave + mortality), + Context (adds metro status and county mask mandates), and + Demographics (adds sex, race, age, and education). Sample restricted to Democrats and Republicans.

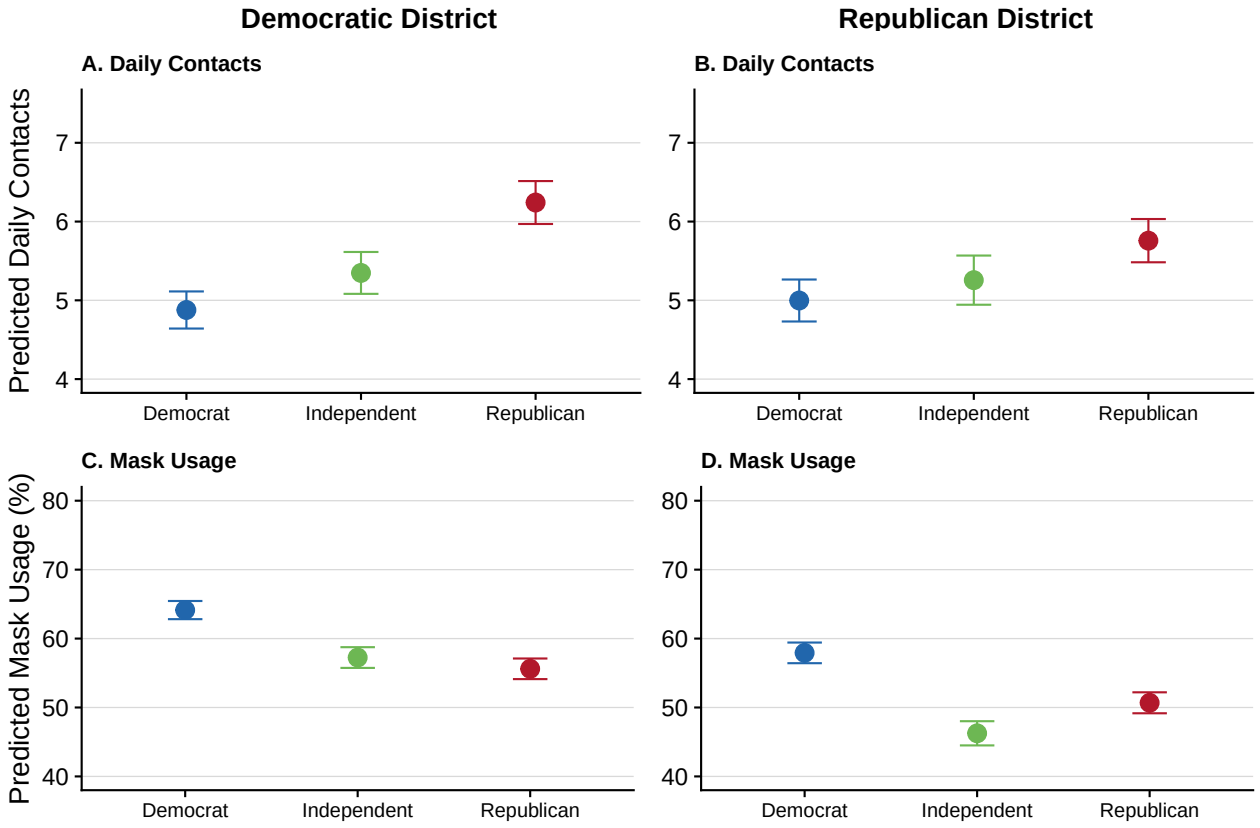


Figure 25: Predicted Health Behaviors by Party and Congressional District Context. Panels A-B show predicted daily contacts; Panels C-D show predicted mask usage. Left column shows Democratic-led districts; right column shows Republican-led districts. Estimates are adjusted for local incidence, mask mandates, metro status, and demographics. Partisan gaps persist in both district contexts.

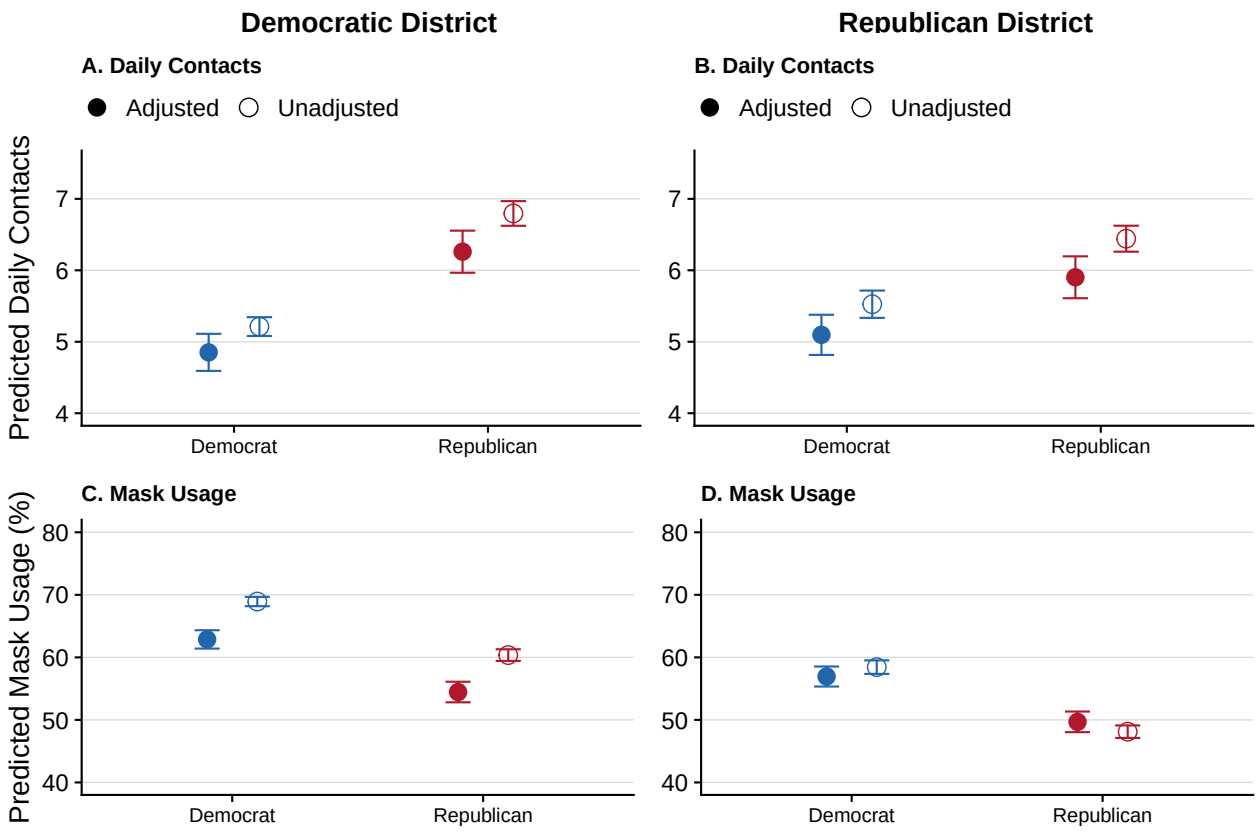


Figure 26: Robustness of Partisan Gaps to Model Specification. Panels A-B show predicted daily contacts; Panels C-D show predicted mask usage. Left column shows Democratic-led districts; right column shows Republican-led districts. Open circles show unadjusted estimates (wave only); filled circles show adjusted estimates (controlling for incidence, mask mandates, metro status, demographics). Partisan gaps persist regardless of model specification.

in Republican districts, the gap remains approximately 0.5-0.8 contacts. For masks, the Democratic advantage over Republicans remains stable at approximately 7-10 percentage points across both district contexts and model specifications. This stability indicates that partisan differences in health behaviors are not explained by compositional differences or local contextual factors.

7.8 Weight Sensitivity Analysis

To assess the robustness of our findings to different weighting approaches, we compare results across three scenarios: (1) unweighted estimates, (2) original demographic weights that adjust for age, gender, race/ethnicity, and education, and (3) raked weights that additionally adjust for party identification to match national targets from Gallup polling (30% Democrat, 29% Republican, 39% Independent; Gallup (2021)). The raking procedure adjusts for potential correlations between party identification and demographic characteristics in our sample.

A key finding from this analysis is that raked weights and original demographic weights produce identical within-party estimates for all outcomes. This equivalence occurs because the raking procedure applies a constant multiplier to all observations within each party group. When calculating within-party means or party-specific regression coefficients, this constant multiplier cancels out. Raking only matters for population-level estimates that pool across partisan groups, where it adjusts the relative contribution of each party to the overall estimate.

We observe small differences between unweighted and weighted estimates. The Republican-Democrat contact gap is 1.33 contacts per day using unweighted data compared to 1.25 contacts per day with demographic weights—a 6% reduction. The vaccine acceptance gap is -18.6 percentage points (unweighted) versus -17.0 percentage points (weighted)—a 9% reduction in the absolute gap. Mask usage differences remain stable across all weighting scenarios, with the Republican-Democrat gap at approximately -10.8 percentage points regardless of weights used.

Importantly, all partisan differences remain highly significant ($p < 0.001$) across all three weighting scenarios. These results demonstrate that our findings are robust to the choice of weighting specification and that the substantive conclusions about partisan gaps in health behaviors do not depend on how we adjust for sample composition.

7.8.1 Summary Table

Table 2: Partisan Differences in I

Outcome	Comparison	Unweighted	Original Weights
Contacts	Republican - Democrat	1.33 (0.17)***	1.25 (0.17)***
Contacts	Independent - Democrat	0.28 (0.18)NA	0.39 (0.17)*
Mask Usage	Republican - Democrat	-10.79 (0.90)***	-10.78 (0.92)***
Mask Usage	Independent - Democrat	-11.72 (0.99)***	-11.79 (1.00)***
NA	Republican - Democrat	-0.19 (0.02)***	-0.17 (0.02)***
NA	Independent - Democrat	-0.14 (0.02)***	-0.13 (0.02)***

Note: Estimates shown with standard errors in parentheses. Significance: * $p < 0.05$, ** $p < 0.01$, ***

7.8.2 Comparison of Partisan Gaps

7.9 Geographic Variation in Incidence Responsiveness

The main text presents models that pool across survey waves, using temporal variation in pandemic conditions (alongside geographic variation) to identify behavioral responses. To isolate the role of geographic variation, we estimate separate models for each survey wave. Within a single wave, variation in county-level incidence rates is purely geographic, reflecting cross-sectional differences in local pandemic severity rather than temporal trends. This approach allows us to examine whether partisan differences in responsiveness to local conditions are consistent across different periods of the pandemic.

Table 3 presents the estimated change in daily contacts associated with a 10% increase in

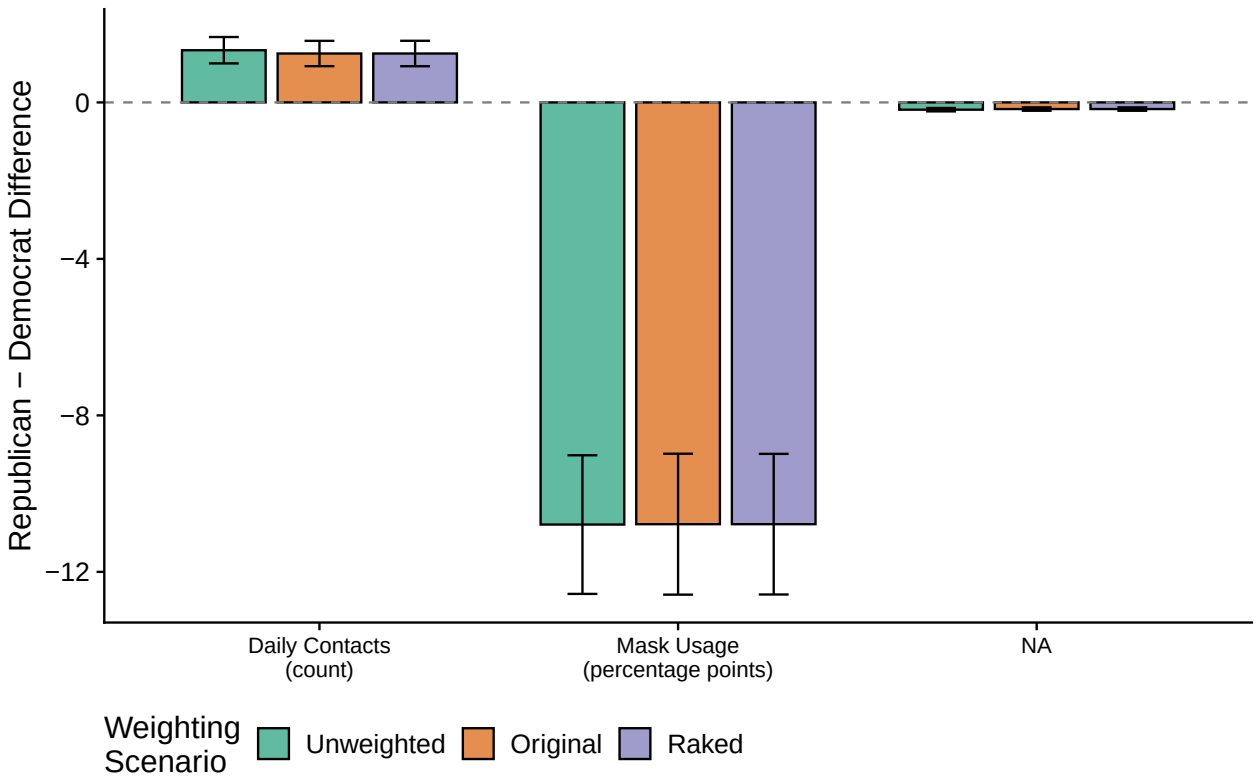
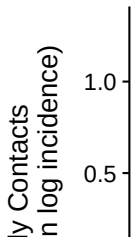


Figure 27: Republican-Democrat Differences in Health Behaviors Across Weighting Scenarios

local (log) incidence rates, by party and wave. All models include state fixed effects to control for time-invariant state characteristics. Figure 28 visualizes these results, showing how the incidence-responsiveness of each partisan group varied over the study period.

Wave	N	Mean Inc. Rate	Democrat	Independent	Republican
Apr '20	2355	131.9	-0.043	-0.017	0.117
Jun '20	2180	51.4	0.249	0.055	-0.007
Sep '20	3040	59.3	-0.369	-0.881	-0.975
Dec '20	2837	284.6	-0.105	0.788	0.079
Feb '21	2827	283.0	0.156	-0.158	-0.118
May '21	4994	72.4	-0.101	0.275	-0.282

Note: Effects represent the change in daily contacts associated with a 10% increase in local log incidenc



7.9.1 Interpretation

The wave-by-wave analysis reveals several patterns. First, Republicans consistently show stronger contact reductions in response to local incidence than Democrats across most survey waves. In 4 of 6 waves, the Republican-Democrat gap change is negative, indicating that higher local incidence narrows the partisan contact gap. Second, the magnitude of responsiveness varies over time, with the strongest Republican responses occurring during Sep '20 when mean incidence was 59.3 cases per 100,000.

These within-wave analyses provide complementary evidence to the pooled models in the main text. By isolating geographic variation in incidence, we demonstrate that partisan differences in responsiveness are not artifacts of differential exposure to temporal trends in the pandemic. Rather, when comparing individuals facing different local conditions at the same point in time, Republicans are more responsive to higher local incidence than Democrats—a pattern that helps explain the narrowing of partisan gaps during periods of elevated disease burden.

7.10 Partisan Behavior Levels Over Time

The main text documents how partisan gaps in health behaviors varied over the course of the pandemic. To understand what drives these fluctuations, Figure 29 displays the average behavior levels for each party across survey waves. This decomposition reveals an important asymmetry: **Democrats maintained relatively stable behavior throughout the pandemic, while Republicans showed substantially greater variability**.

For contact behavior (Panel A), Democrats maintained a relatively narrow range of daily contacts across all six waves, while Republicans exhibited pronounced swings—increasing contacts substantially during periods of lower case counts (September 2020) and reducing them during surges (December 2020, February 2021). The coefficient of variation for Republican contacts (25.7%) was notably higher than for Democrats (15.8%), confirming greater Republican variability.

Mask usage (Panel B) shows a different pattern. All parties increased mask-wearing from

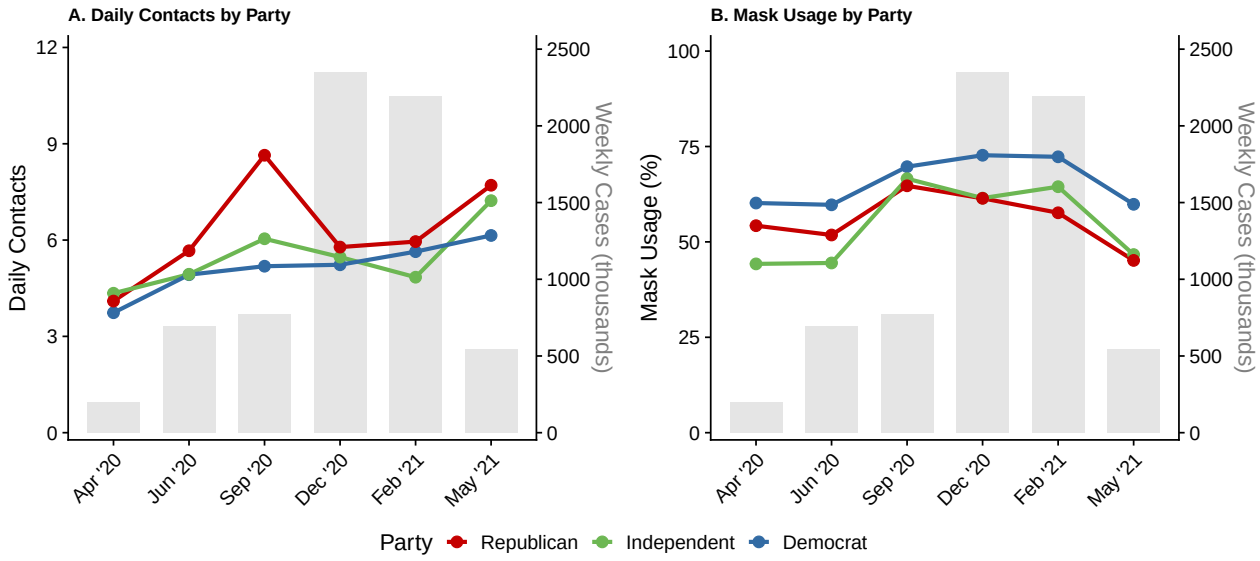


Figure 29: Health Behavior Levels by Party Across Survey Waves. Panel A shows mean daily contacts; Panel B shows mask usage (percentage of contacts with masks). Gray background bars indicate weekly COVID-19 cases (in thousands). Democrats (blue) maintained relatively stable behavior while Republicans (red) showed greater variability, particularly in contact behavior.

the early pandemic through winter 2020–21, but Republicans started from a lower baseline and showed steeper declines as vaccines became available in spring 2021. Democrats maintained consistently high mask usage throughout, while Republican mask-wearing tracked more closely with pandemic conditions. This asymmetry—stable Democratic behavior paired with variable Republican behavior—is the primary driver of the fluctuating partisan gaps documented in the main text.

7.10.1 Formal Test of Variance Differences

To formally test whether variance in contact behavior differs across partisan groups, we conducted Levene's test for equality of variances on individual-level contact data. The results confirm significant heterogeneity in variance across parties ($F(2, 18{,}230) = 15.47$, $p < 0.001$). Republicans exhibited the highest variance in daily contacts (120.3), followed by Independents (107.9) and Democrats (83.0). Table 4 summarizes these results.

Table 4: Variance in Daily Contacts by Political Party with Levene's T

Party	N	Mean Contacts	SD
Republican	5517	6.87384448069602	10.9667090980222
Independent	4527	5.82372432074218	10.3894697106189
Democrat	8189	5.53938209793628	9.10909661236962
Levene's Test	NA	NA	F = 15.47

Note: Levene's test assesses equality of variances across groups. Significant result ($p < 0.001$) indicates

The significantly higher variance among Republicans indicates that Republican contact behavior was more responsive to changing conditions—whether pandemic severity, policy environment, or media messaging—while Democrats maintained more consistent behavior throughout the study period.

7.11 Detailed Pandemic Response by Party

Figure 30 shows the standardized regression coefficients for the relationship between prior week's pandemic conditions (incidence and mortality) and health behaviors (contacts and mask usage), by party. Effects are standardized: bars show the change associated with a one standard deviation increase in log incidence or log mortality.

Figure 31 summarizes how partisan gaps change in response to pandemic conditions. This figure complements the individual-level effects shown above by directly displaying the implications for polarization.

7.11.1 Incidence vs. Mortality: Why We Focus on Mortality

The main text focuses on mortality rather than incidence when discussing partisan responses to pandemic conditions. This section explains that choice and presents the full comparison of incidence and mortality effects.

Key finding: Mortality was the stronger predictor of contact behavior, while incidence was the stronger predictor of mask-wearing.

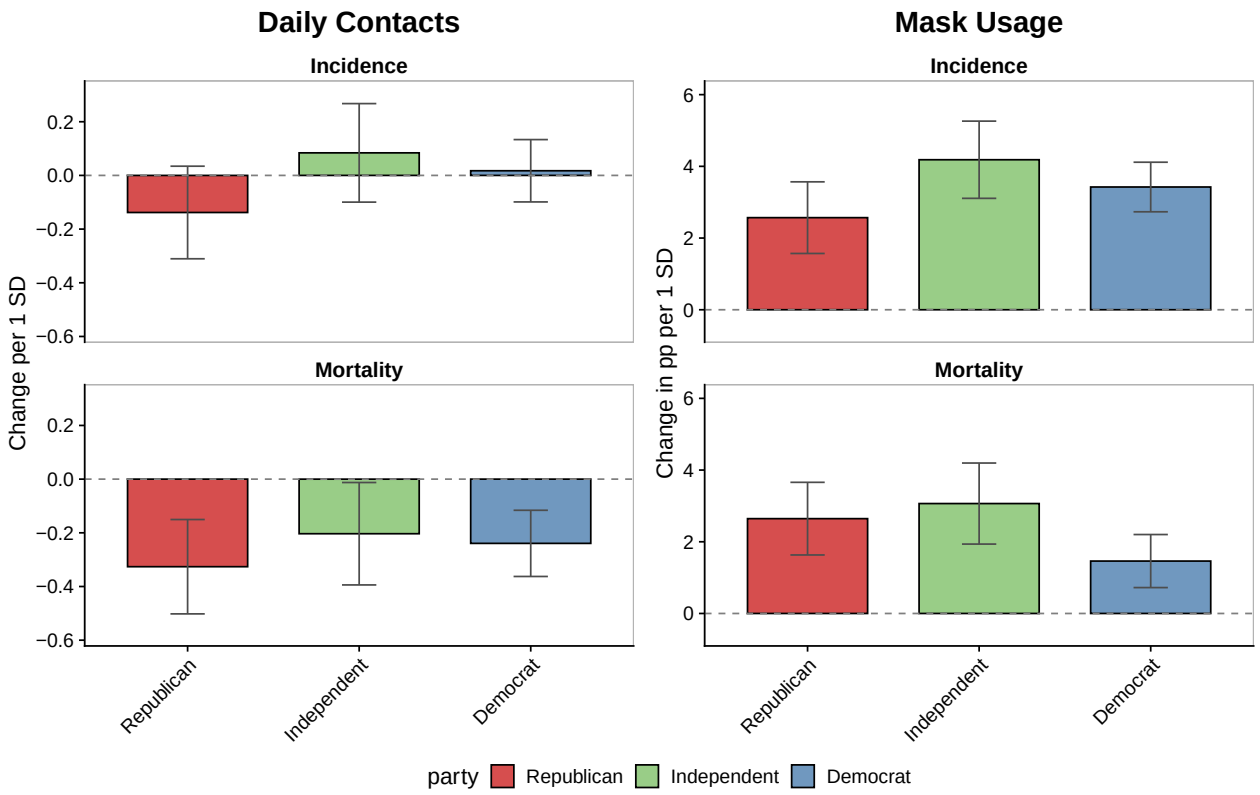


Figure 30: Detailed Behavioral Response to Prior Week's Pandemic Conditions by Party. Left column shows changes in daily contacts; right column shows changes in mask usage (percentage points). Top row shows response to incidence; bottom row shows response to mortality. Effects are standardized: bars show the change associated with a one standard deviation increase in log incidence or log mortality. Models include county fixed effects. Error bars indicate standard errors. Negative values for contacts indicate reductions; positive values for masks indicate increases.

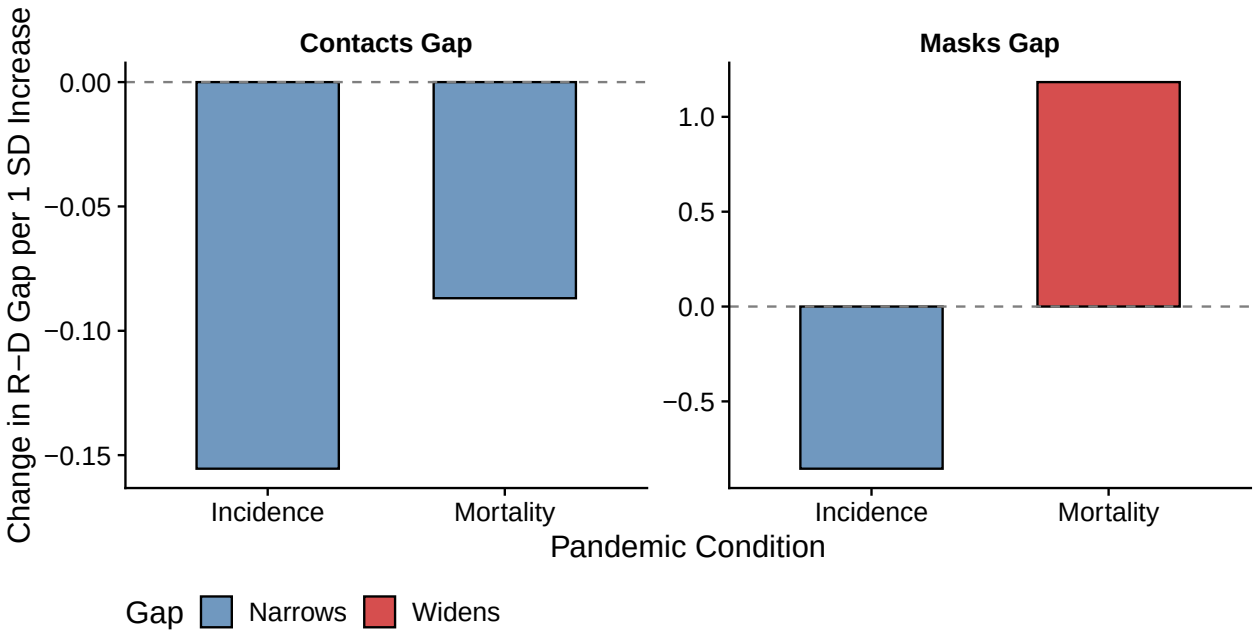


Figure 31: Effect of Prior Week's Pandemic Conditions on Partisan Gaps. Bars show how the Republican-Democrat gap changes in the week following a one standard deviation increase in local log incidence or mortality. Negative values indicate gap narrowing (convergence); positive values indicate gap widening (divergence).

Figure 30 above displays the full comparison. For contacts (left column), mortality effects are larger than incidence effects for all partisan groups, and statistically significant for Republicans. For masks (right column), the pattern reverses: incidence effects are larger, particularly for Democrats.

This differential pattern has a plausible interpretation:

1. **Mortality and contacts**: Deaths are concrete, visible, and emotionally salient. They are reported in local news, experienced through social networks, and marked by funerals and obituaries. When deaths rise, the threat becomes tangible—prompting reductions in social contact. Republicans showed the largest mortality response, consistent with the main text's finding that their contact behavior swung more dramatically with conditions.

2. **Incidence and masks**: Case counts, by contrast, are more abstract and mediated by testing availability and reporting. However, rising case counts may trigger mask mandates, workplace policies, and social pressure to mask—particularly in areas where

cases are visibly surging. Democrats, who were more likely to live in areas with mask mandates and to follow public health guidance, showed the largest incidence response for masks.

Figure 31 summarizes the implications for polarization: mortality narrows the contact gap (Republicans reduce contacts more), while incidence slightly widens the mask gap (Democrats increase masking more). This supports the main text's focus on mortality for the contact behavior analysis, where the key finding is that objective threat can reduce polarization—at least temporarily.

The sections below provide additional detail on mask responses to incidence (Section 7.11.2) and perception responses to both conditions (Section 7.11.3).

7.11.2 Mask Usage Response to Incidence

Figure 32 parallels Figure 6 in the main text but shows mask usage response to incidence rather than contacts response to mortality. This figure illustrates the finding that incidence was the stronger predictor of mask-wearing, particularly for Democrats.

7.11.3 Perception Response to Pandemic Conditions

Figure 33 shows how COVID-19 concern (the percentage reporting being "very concerned") responded to prior week's pandemic conditions by party. This parallels Figure 30 but examines perceptions rather than behaviors.

Unlike health behaviors, which showed clear differential responses to pandemic conditions (mortality predicting contacts, incidence predicting masks), risk perception showed weaker and more inconsistent relationships with local pandemic conditions. This suggests that partisan differences in concern were more stable and less responsive to objective signals about pandemic severity—consistent with the finding in the main text that perception gaps remained relatively constant even as behavior gaps fluctuated.

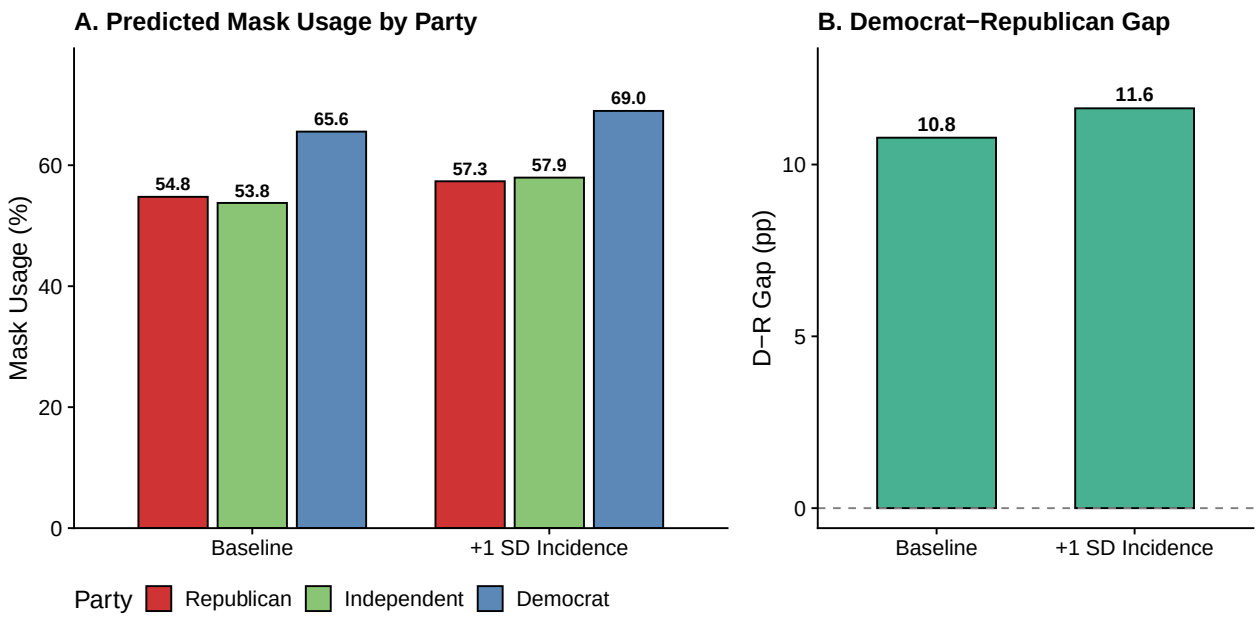


Figure 32: Mask Usage Response to Incidence. Panel A shows predicted mask usage (percentage of contacts with masks) by party at baseline (average incidence) and when incidence is one standard deviation higher. Panel B shows the Democrat-Republican gap in mask usage at baseline and at higher incidence. All partisan groups increase mask usage when incidence rises, with Democrats showing the largest response.

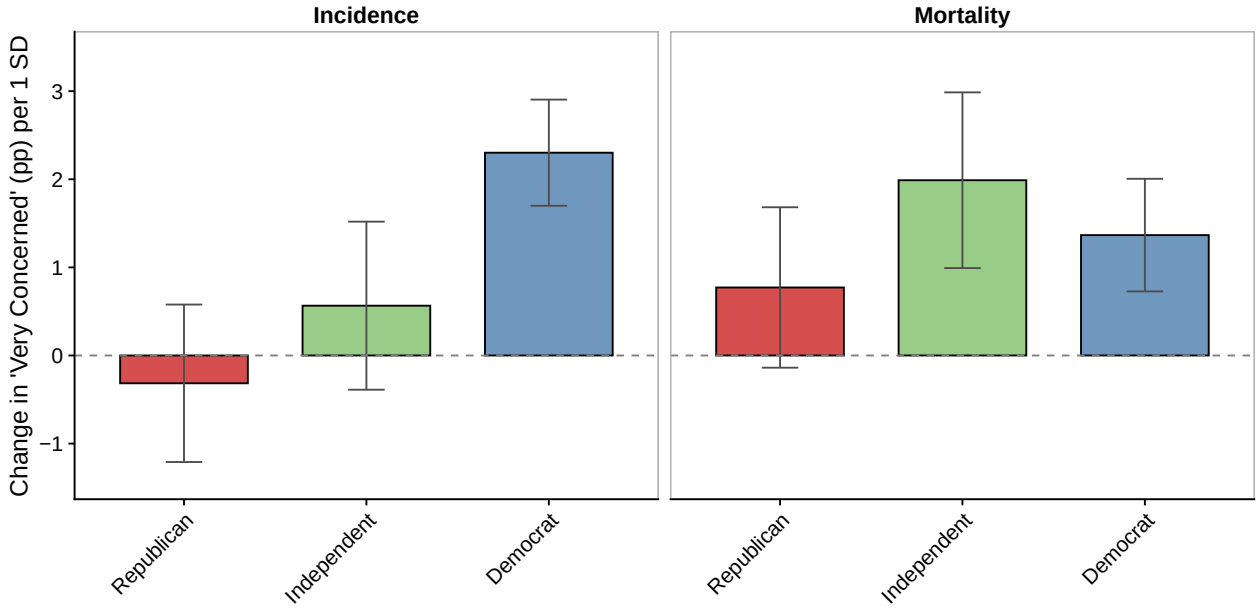


Figure 33: Risk Perception Response to Prior Week's Pandemic Conditions by Party. Left column shows response to incidence; right column shows response to mortality. Bars show the change in the percentage reporting being 'very concerned' about COVID-19 associated with a one standard deviation increase in log incidence or log mortality. Models include county fixed effects. Error bars indicate standard errors.

7.12 Spatial Variation in Partisan Differences

This section provides additional detail on geographic variation in partisan differences. All estimates in this section are adjusted for survey wave and local mortality rates using regression models, removing potential confounding from differences in when surveys were collected and local pandemic severity. We present behavior levels by party within census regions (complementing the region-level gap maps in the main text), then examine finer geographic scales: census divisions (9 divisions) and states.

Figure 34 displays adjusted health behavior levels by party and census region.

7.12.1 Urban-Rural Behavior Levels by Party

Figure 35 displays the Republican-Democrat gap in contacts and mask-wearing by metropolitan status. The gaps are smaller in non-metropolitan areas for both behaviors: the contact gap drops from 1.3 to 0.3 additional daily contacts, and the mask gap drops from 9 to 7 percentage points. This pattern helps explain why the South—which has a higher proportion of non-metropolitan residents—showed smaller partisan gaps in the main analysis.

Figure 36 displays health behavior levels by party and metropolitan status, showing that all partisan groups were less cautious in non-metropolitan areas.

The figure reveals that the urban-rural shift toward less protective behavior is consistent across all three partisan groups. Notably, Independents show the largest urban-rural differences, particularly for mask-wearing and vaccine acceptance.

7.12.2 Partisan Gaps Within Policy Environments

A potential confounder for geographic patterns in partisan gaps is that metropolitan areas were more likely to have strict mask mandates. Figure 37 shows that partisan gaps persist within each policy environment, indicating that individual partisanship shapes behavior beyond policy effects.

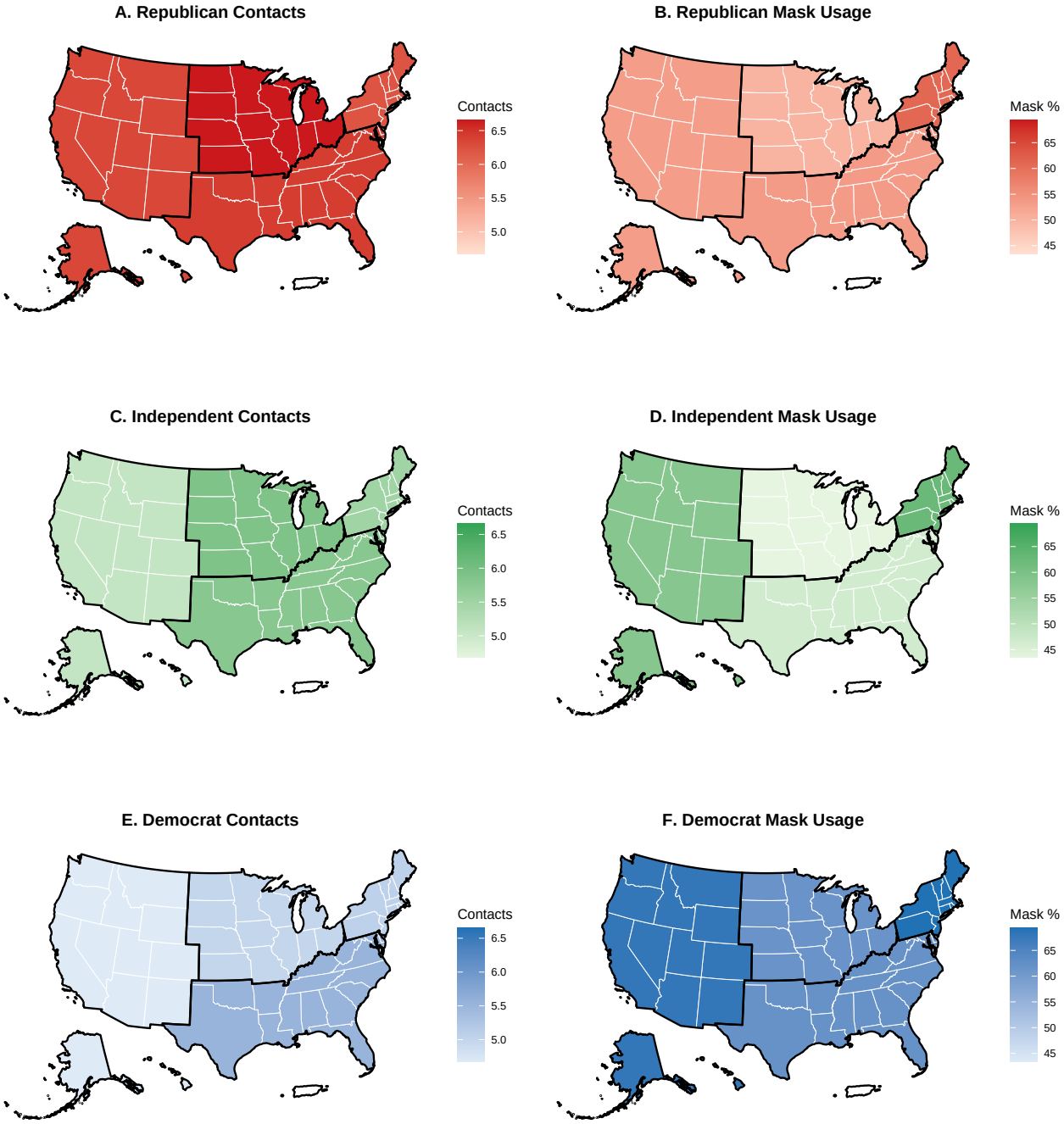


Figure 34: Adjusted Health Behavior Levels by Party Across Census Regions. Each row shows a partisan group: Republicans (A-B), Independents (C-D), and Democrats (E-F). Left column shows mean daily contacts; right column shows mask usage. Estimates are adjusted for survey wave and local mortality rates. Darker shading indicates higher values. Black borders indicate census region boundaries.

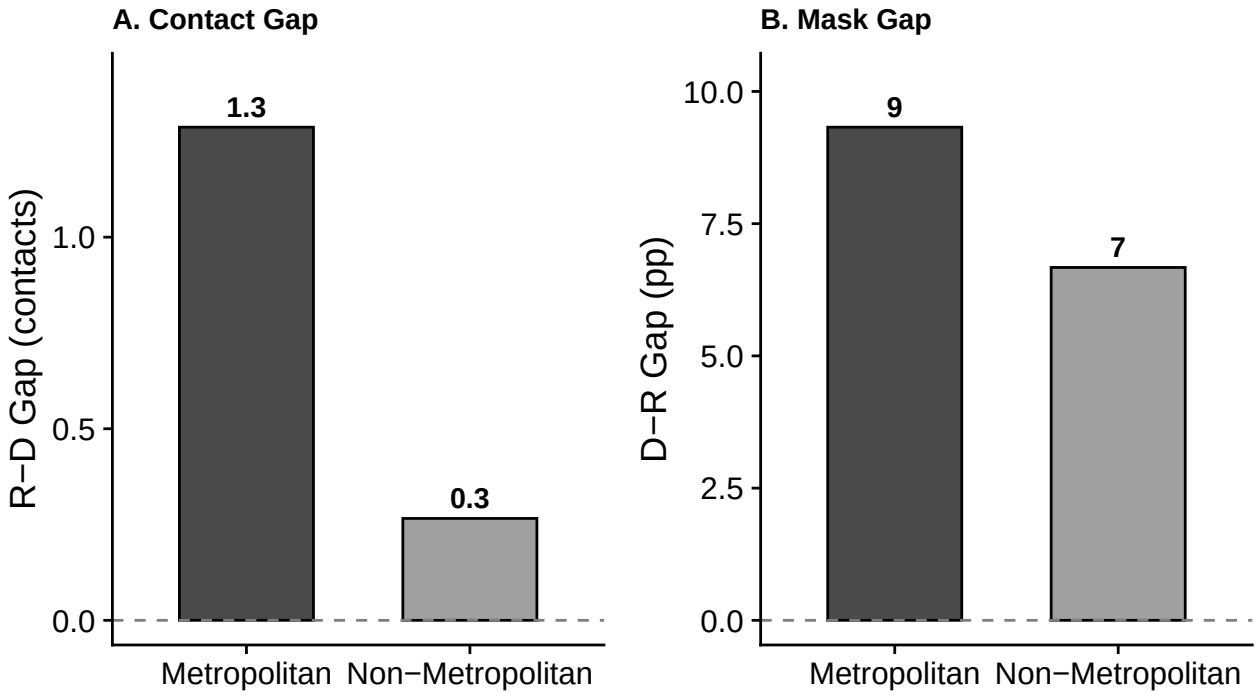


Figure 35: Republican-Democrat Gaps by Metropolitan Status. Each panel compares the partisan gap in metropolitan versus non-metropolitan areas. Panel A shows the contact gap (R−D) and Panel B shows the mask gap (D−R). Estimates are adjusted for survey wave and local mortality rates.

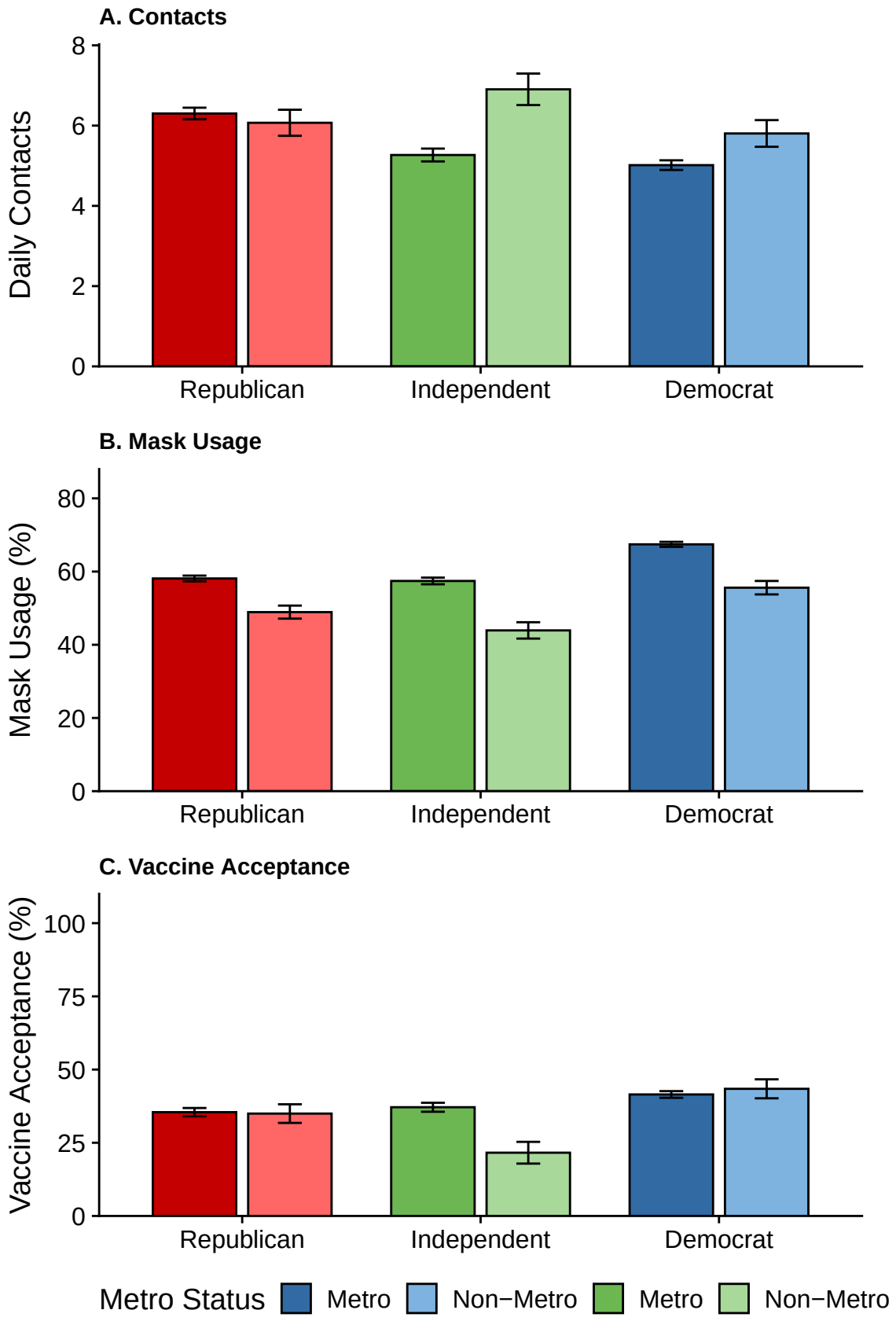


Figure 36: Adjusted Health Behaviors by Party and Metropolitan Status. Each panel shows behavior levels for the three partisan groups, comparing metropolitan (darker bars) versus non-metropolitan (lighter bars) areas. Panel A shows daily contacts, Panel B shows mask usage, and Panel C shows vaccine acceptance rates (Wave 6 only). Estimates are adjusted for survey wave and local mortality rates. Error bars indicate standard errors.

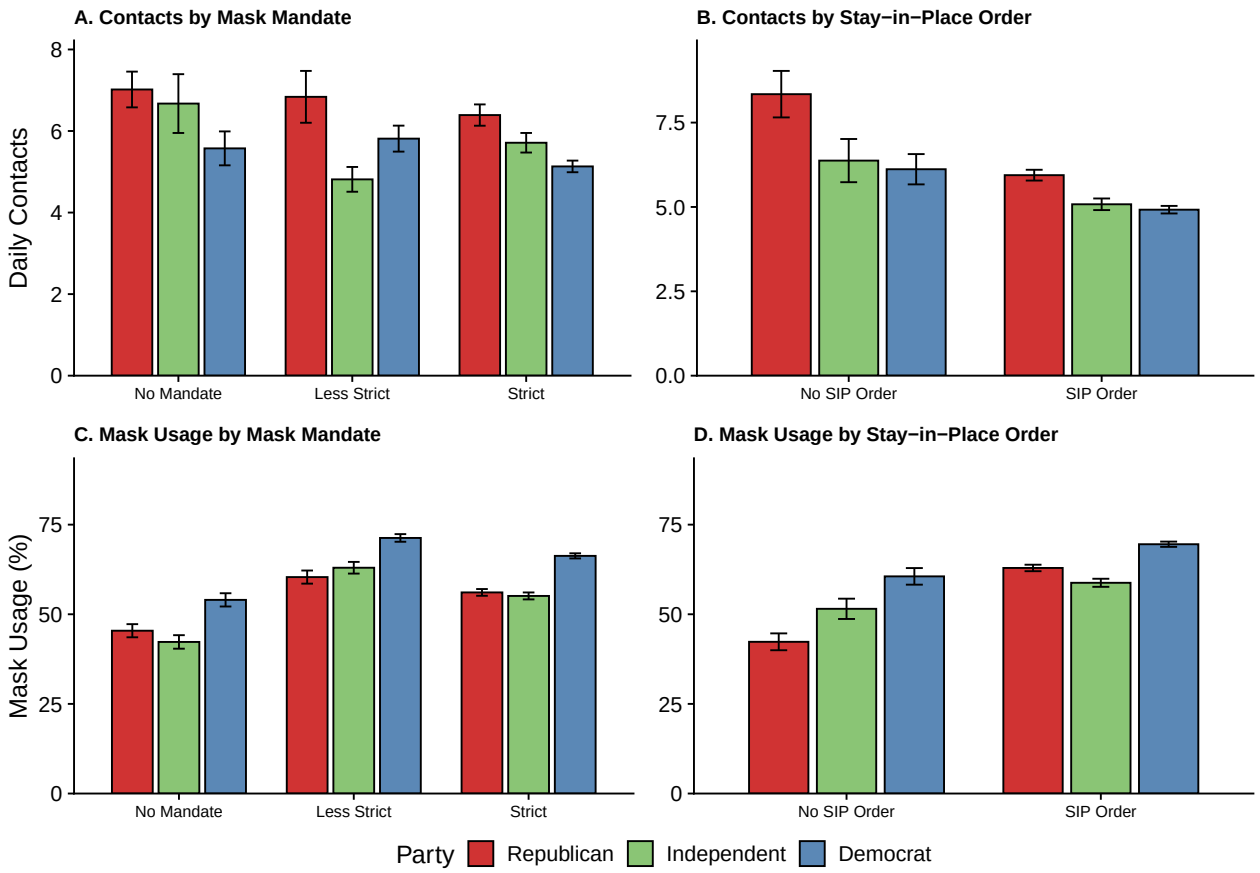


Figure 37: Health Behaviors by Party and Policy Environment. Left column shows behavior by county mask mandate policy (no mandate, less strict, strict). Right column shows behavior by respondent-reported stay-in-place order status (waves 1-5 only; this question was not assessed in wave 6). Top row shows daily contacts; bottom row shows mask usage. Partisan gaps persist within each policy environment.

7.12.3 Behavior by Local Partisan Context

Figure 38 shows daily contacts by party and local partisan context (congressional district representation). The figure reveals that partisan gaps in contacts persist regardless of local context, though the magnitude varies.

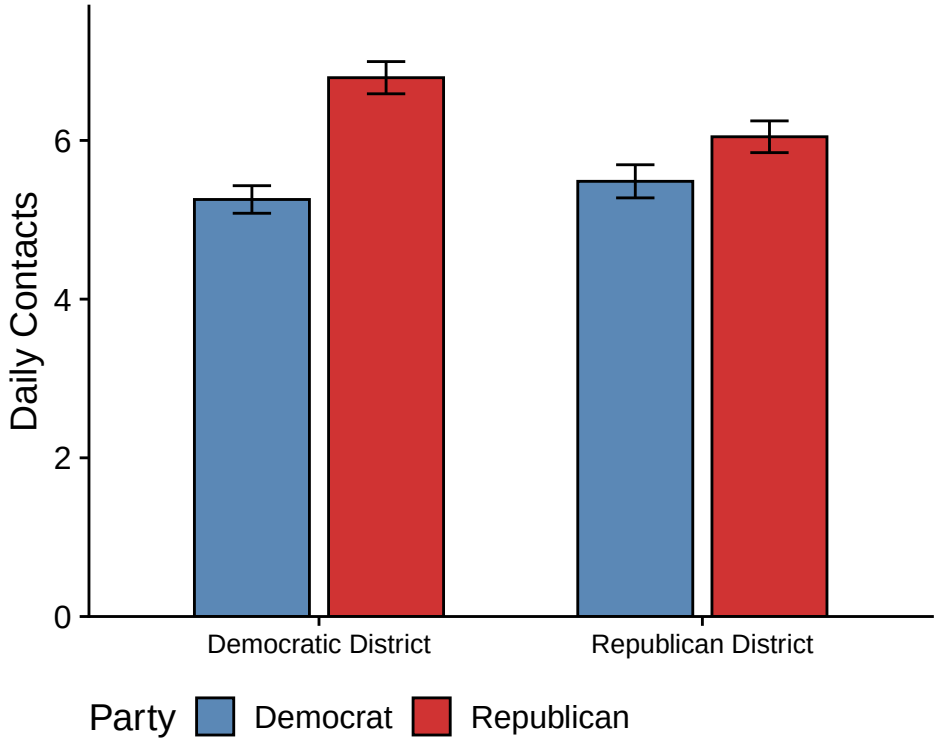


Figure 38: Daily Contacts by Party and Local Partisan Context. Bars show mean daily contacts for Democrats and Republicans depending on whether they live in a Democratic-led or Republican-led congressional district. Estimates are adjusted for survey wave, local incidence rates, and metropolitan status. Error bars indicate standard errors.

Figure 39 shows the Republican-Democrat gap in each behavior by local partisan context. For contacts, the R-D gap is larger in Democratic-led districts than in Republican-led districts, suggesting that Republicans increase contacts more than Democrats when living in Democratic areas. For masks, the D-R gap is similar across contexts, indicating that both parties shift their mask behavior in parallel when living in opposing-party areas.

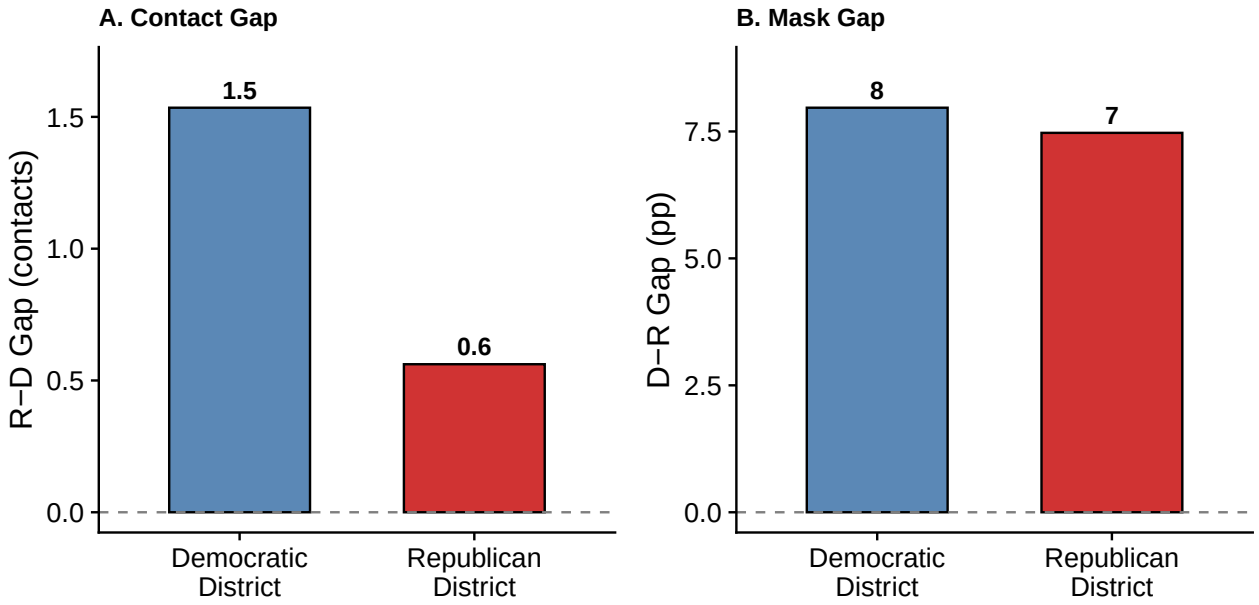


Figure 39: Partisan Gaps by Local Context. Panel A shows the Republican-Democrat gap in daily contacts; Panel B shows the Democrat-Republican gap in mask usage. Gaps are calculated within each congressional district context (Democratic-led vs Republican-led). Estimates are adjusted for survey wave, local incidence rates, and metropolitan status.

7.12.4 Regional Behavior Levels

Figure 40 displays the raw behavior levels (mean contacts and mask usage) for each party within each census region. This complements the gap-focused maps in the main text by showing the underlying distributions.

The figure reveals that the smaller partisan gaps in the South are driven primarily by Republicans in that region reporting fewer contacts and higher mask usage than Republicans elsewhere, rather than by Democrats in the South behaving differently. This suggests that regional norms may exert a moderating influence on Republican behavior specifically.

7.12.5 Regional Vaccine Acceptance and Perception Maps by Party

Figure 41 displays vaccine acceptance rates (Wave 6 only) and pandemic concern levels by party across census regions. These complement the contact and mask usage maps above.

Vaccine acceptance rates show strong partisan gaps across all regions, with Democrats consistently reporting higher acceptance rates than Republicans. The gap is largest in the South

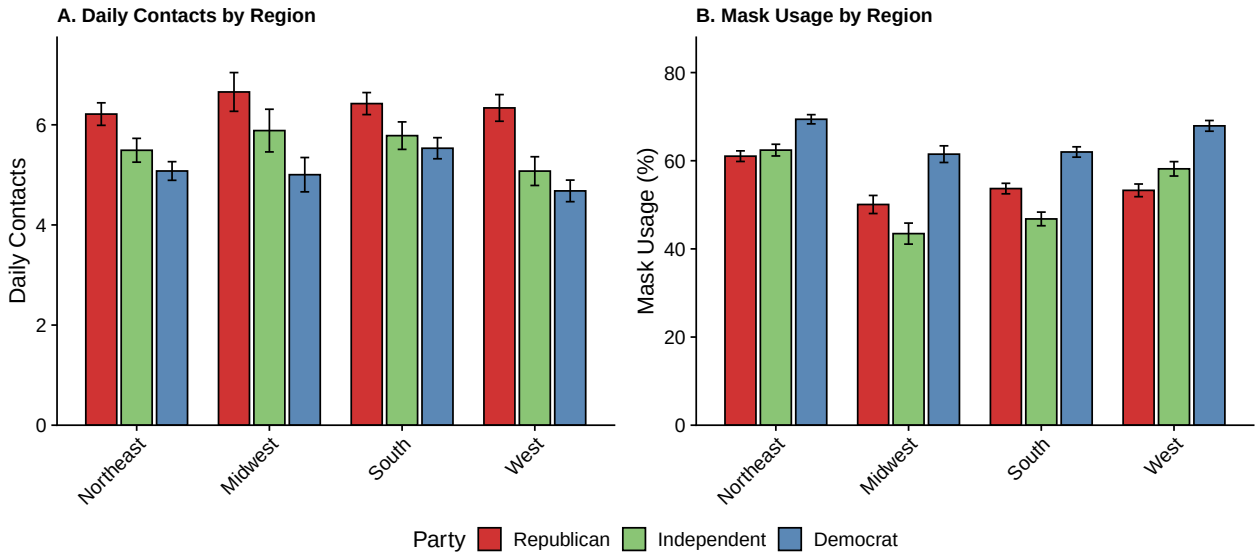


Figure 40: Adjusted Health Behaviors by Party Across Census Regions. Panel A shows mean daily contacts; Panel B shows mask usage (percentage of contacts with masks). Estimates are adjusted for survey wave and local mortality rates. Error bars indicate standard errors from emmeans.

and Midwest. Perception of pandemic severity (concern) shows an even starker partisan divide: Democrats report being "very concerned" about COVID-19 at rates 20-30 percentage points higher than Republicans in every region. Interestingly, while Republican concern varies less across regions, Democratic concern is highest in the Northeast and West.

7.12.6 All Partisan Gaps by Census Region

Figure 42 displays partisan gaps across all four outcomes in a single 2×2 view, facilitating comparison of geographic patterns across behaviors and perceptions.

The 2×2 view reveals that geographic patterns differ across outcomes. Contact gaps (A) and mask gaps (B) show the most regional variation, with the South having notably smaller gaps. In contrast, vaccine acceptance gaps (C) and especially concern gaps (D) are more uniform across regions, suggesting that these outcomes are more strongly driven by partisanship itself rather than regional context.

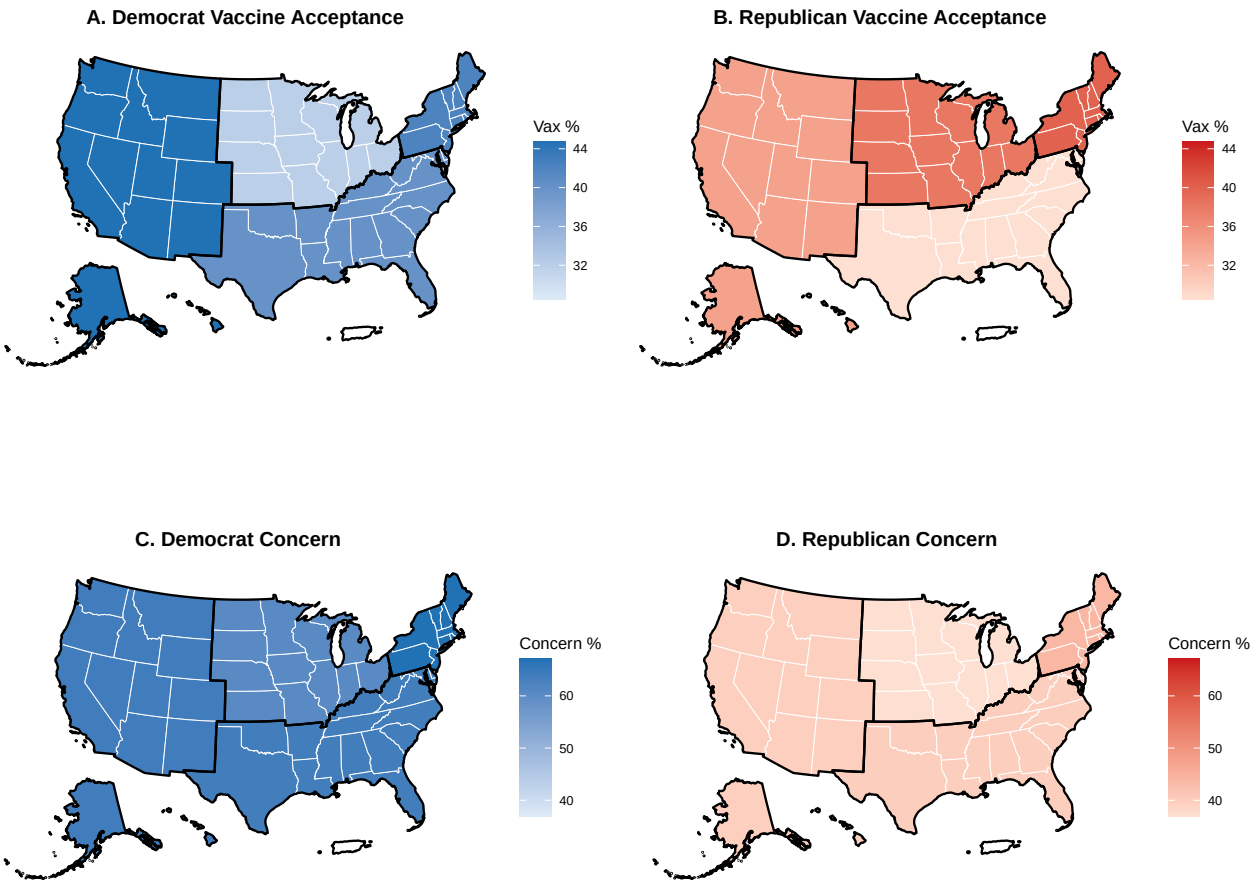


Figure 41: Adjusted Vaccine Acceptance and Perception Levels by Party Across Census Regions. Top row shows vaccine acceptance rates (Wave 6, May 2021) for Democrats (A) and Republicans (B). Bottom row shows percentage reporting being 'very concerned' about COVID-19 for Democrats (C) and Republicans (D). Estimates are adjusted for local mortality rates. Darker shading indicates higher values. Black borders indicate census region boundaries.

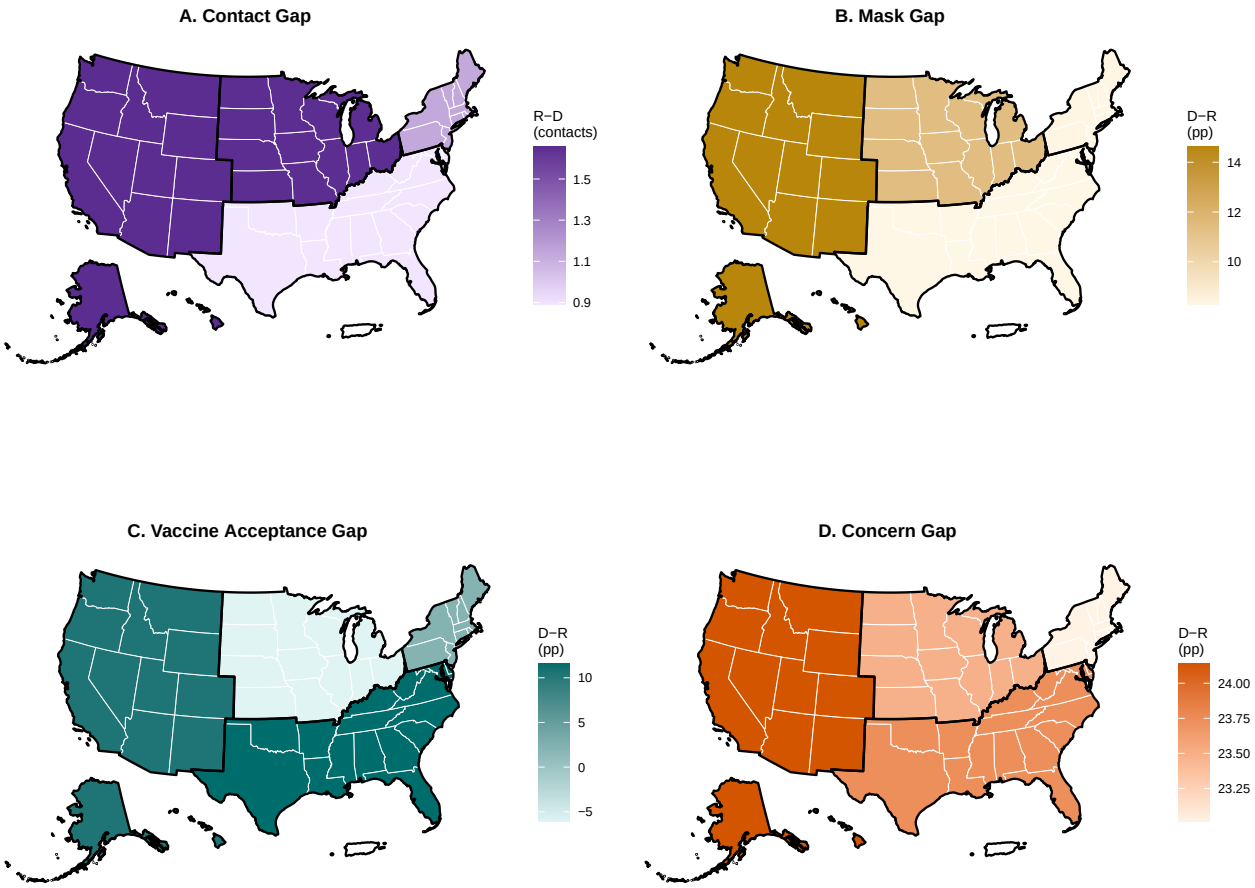


Figure 42: Geographic Distribution of All Partisan Gaps by Census Region (Adjusted). Panel A shows the Republican-Democrat gap in daily contacts (purple); Panel B shows the Democrat-Republican gap in mask usage (gold); Panel C shows the Democrat-Republican gap in vaccine acceptance (teal); Panel D shows the Democrat-Republican gap in COVID-19 concern (orange). Estimates are adjusted for survey wave and local mortality rates. Black borders indicate census region boundaries.

7.12.7 Partisan Gaps by Census Division

Figure 43 displays partisan gaps at the census division level (9 divisions), providing more geographic granularity than the 4 census regions. **Note**: The East South Central division (AL, KY, MS, TN) shows a reversed pattern with Democrats reporting more contacts and higher mask usage than Republicans. This anomaly likely reflects sampling variability given the smaller sample sizes in that division (n=166 Democrats, n=205 Republicans).

7.12.8 State-Level Partisan Gaps

Figure 44 provides a more granular view, mapping state-level partisan differences in contacts and mask usage. **Important caveat**: The BICS sample was designed to be nationally representative, not representative at the state level. State-level estimates should therefore be interpreted with caution, as they may reflect sampling variability rather than true geographic differences. We present these maps for exploratory purposes only; the regional and division-level analyses above provide more reliable geographic comparisons. States with fewer than 10 respondents per party are excluded.

The maps reveal substantial heterogeneity in partisan gaps across states. For contact behavior, most states show the expected pattern of Republicans reporting more contacts than Democrats (red shading), but the magnitude varies considerably—from nearly no gap in some Southern states to gaps exceeding 5 contacts per day in states like Michigan and Colorado. A few states show reversed patterns (blue shading) where Democrats report more contacts, though these often reflect small sample sizes.

For mask usage, the Democrat-Republican gap (blue shading indicates Democrats masking more) is positive in nearly all states, but again varies substantially in magnitude. Western states tend to show larger mask gaps, while some Southern states show smaller differences—consistent with the regional patterns documented in the main text.

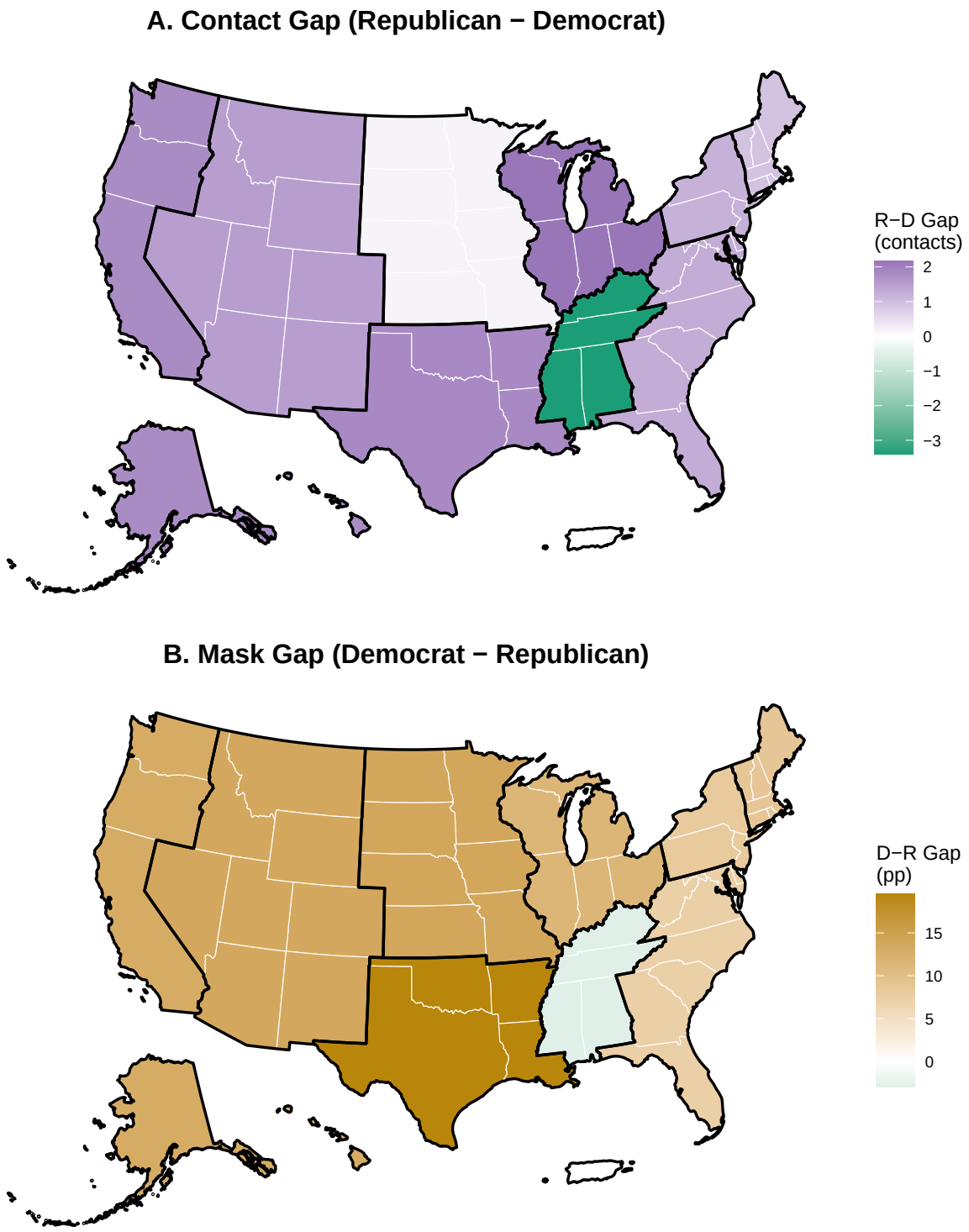


Figure 43: Geographic Distribution of Adjusted Partisan Gaps by Census Division. Panel A shows the Republican-Democrat gap in daily contacts (purple = Republicans report more contacts; teal = Democrats report more). Panel B shows the Democrat-Republican gap in mask usage (gold = Democrats mask more; teal = Republicans mask more). Estimates are adjusted for survey wave and local mortality rates. White indicates no gap. Black borders indicate census division boundaries.

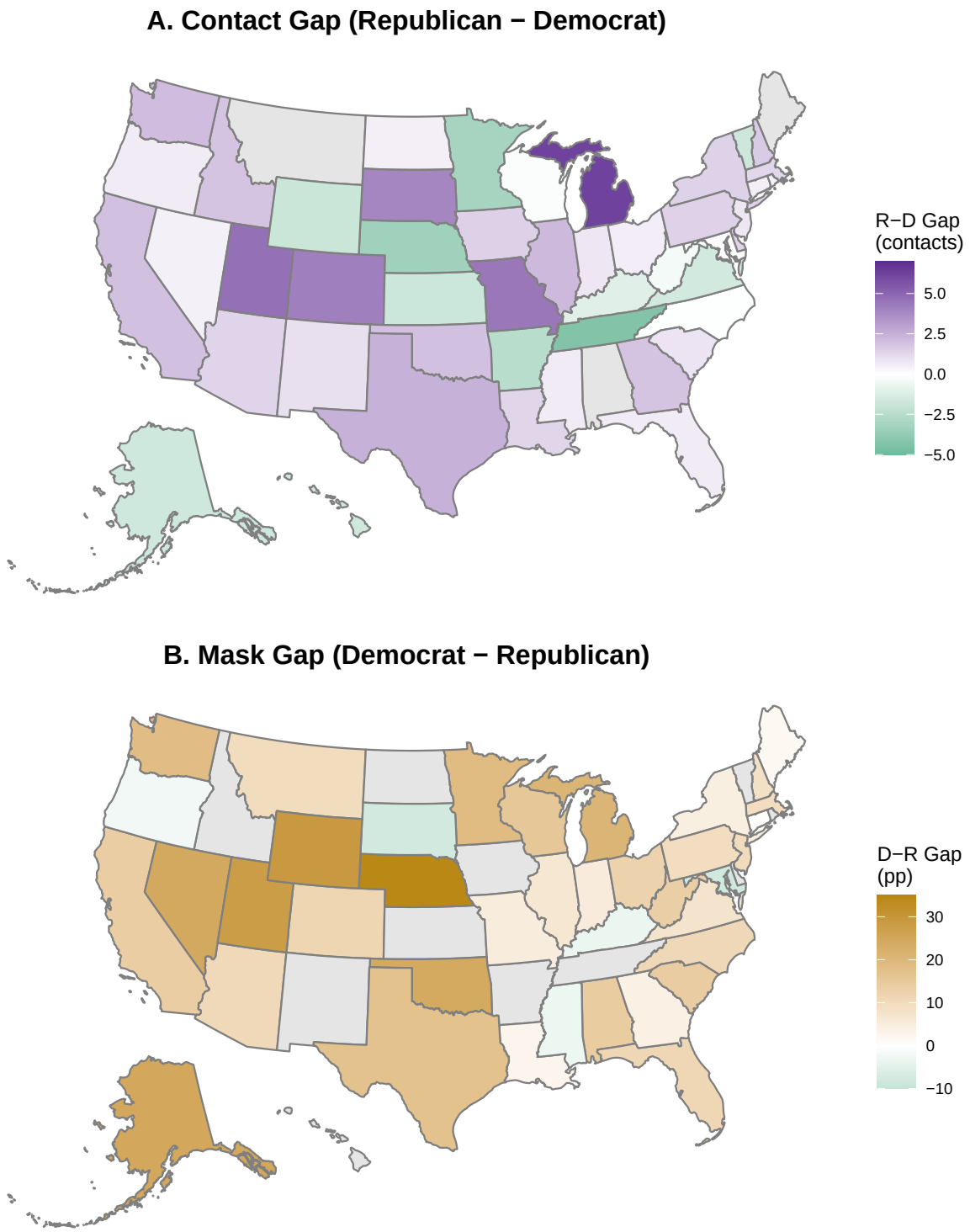


Figure 44: State-Level Adjusted Partisan Gaps in Health Behaviors. Panel A shows the Republican-Democrat gap in daily contacts (purple = Republicans report more; teal = Democrats report more). Panel B shows the Democrat-Republican gap in mask usage (gold = Democrats mask more; teal = Republicans mask more). Estimates are adjusted for survey wave and local mortality rates. White indicates no gap; gray states have insufficient sample size.